

Chapter 10

MAT-130: List of Practice Problems

Please practice following exercises

7.8: 3, 4, 5, 10, 15, 17, 23

10.1: 3, 5, 6, 9, 12, odd: 65-70, odd: 75-80

10.2: 3, 5, 7, 10, 11, 24, 25, 26, 29, 45

10.3: 3, 5, 9, 13, 17, 19, 21, 36, 38

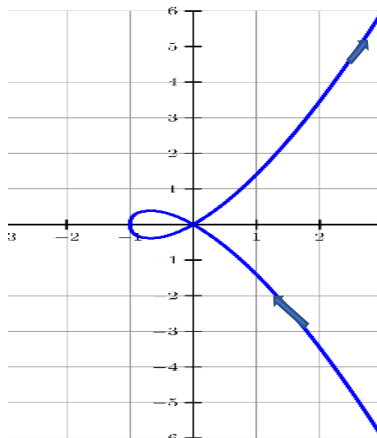
Section 10.1: Parametric Equations

Summary:

1. Graphing Parametric Equations
2. Finding slopes of tangent lines
3. Finding points where the tangent line is either vertical or horizontal
4. Finding length of a parametric curve (6.4)
5. Finding the area of a surface of revolution (6.5)

PARAMETRIC EQUATIONS:

Suppose that a particle moves along a curve C in the xy -plane in such a way that its x – and y –coordinates, as functions of time, are $x = f(t)$, $y = g(t)$. We call these the parametric equations of motion for the particle and refer to C as the trajectory of the particle or the graph of the equations. The variable t is called the parameter of the equations.



$x = t^2 - 1$, $y = t^3 - t$, $-\infty < t < \infty$. The loop comes for t –values from -1 to 1 .

Functions in parametric form: input = parameter, output = $f(\text{parameter})$

$y = f(x)$: $x = t$, $y = f(t)$, The domain of the function is the set of parameters.

$x = f(y)$: $y = \theta$, $x = f(\theta)$, The domain of the function is the set of parameters.

Parametric Equations for Few well-known Graphs

Linear Line:

1. Equations: $x = at + b$, $y = mt + n$, t any real number.

2. For example, a **line segment** that starts from (x_1, y_1) and ends to (x_2, y_2) is parametrized by

$$[x = (\text{initial } x - \text{value}) + (\text{run})t, \quad y = (\text{initial } y - \text{value}) + (\text{rise})t, \quad 0 \leq t \leq 1]$$

$$x = x_1 + (x_2 - x_1)t, \quad y = y_1 + (y_2 - y_1)t, \quad 0 \leq t \leq 1.$$

Parametric Equations of few other curves:

Curve	Parametric Equations
Circle	$x = r \cos t$ $y = r \sin t$ (r is radius)
Ellipse	$x = a \cos(ct)$ $y = b \sin(ct)$ (a, b, c are constants $\neq 0$)
Parabola	vary

Equation of a parabola	Parametric equations
$y^2 = 4ax$	$x = at^2, y = 2at$
$y^2 = -4ax$	$x = -at^2, y = 2at$
$x^2 = 4ay$	$x = 2at, y = at^2$
$x^2 = -4ay$	$x = 2at, y = -at^2$
$(y-h)^2 = 4a(x-k)$	$x = k + at^2, y = h + 2at$

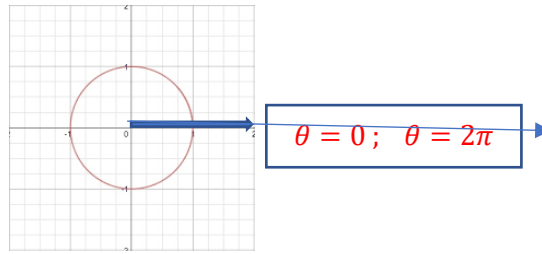
The parametric equations of the circle $x^2 + y^2 = 4^2 \rightarrow x = 4 \cos \theta$, $y = 4 \sin \theta$, $0 \leq \theta \leq 2\pi$.

Equation of a hyperbola	Parametric equations
$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, with centered at origin	$x = a \sec t, y = b \tan t$
$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$, with centered at (h, k)	$x = h + a \sec t, y = k + b \tan t$

Graphing Parametric Curves:

Exercise:1

(a) Find the graph of the parametric equations $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$



Solution:

Given, $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$

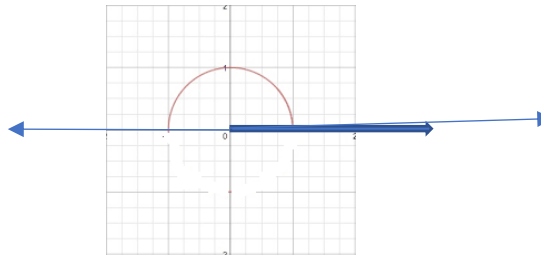
Here we get,

$$x^2 = \cos^2 t, \quad y^2 = \sin^2 t, \quad 0 \leq t \leq 2\pi.$$

That is,

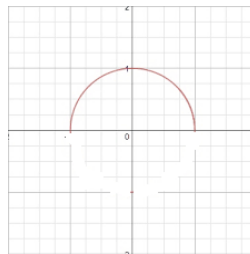
$x^2 + y^2 = 1$, $0 \leq t \leq 2\pi$, which gives us the **whole** circle with center at the (0,0) and radius $r = 1$ in **anticlockwise direction**.

(b) Find the graph of the parametric equations $x = \cos t$, $y = \sin t$, $0 \leq t \leq \pi$.



Orientation is in **anti-clockwise**.

(c) Find the graph of the parametric equations $x = \cos t$, $y = \sin t$, where t moves from $t = \pi$ to $t = 0$.



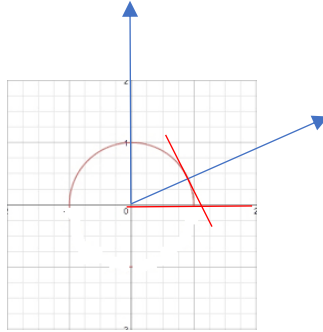
Orientation is **clockwise**.

(d) Find the graph of the parametric equations $x = \cos t$, $y = \sin t$, $\frac{\pi}{6} \leq t \leq \frac{\pi}{2}$.

From $x = \cos t$, $y = \sin t$, we get $x^2 = \cos^2 t$, $y^2 = \sin^2 t$.

Hence, $x^2 + y^2 = \cos^2 t + \sin^2 t = 1$. That is, $x^2 + y^2 = 1$.

The graph of the parametric equations is a piece of the circle $x^2 + y^2 = 1$ with center at the origin and radius 1 from $(\cos \frac{\pi}{6}, \sin \frac{\pi}{6}) = (\frac{\sqrt{3}}{2}, \frac{1}{2})$ to $(\cos 0, \sin 0) = (0, 1)$.



Orientation is in anti-clockwise.

(e) Find the graph of the parametric equations $x = 3 \cos t$, $y = 3 \sin t$, $\frac{\pi}{6} < t \leq \frac{\pi}{2}$

➔ Homework [Write down the solution]

Exercise:2

(a) **Graph** the parametric curve $x = 2t - 3$, $y = 6t - 7$ by eliminating the parameter, and **indicate the orientation** on the graph.

Solution. To eliminate the parameter, we will solve the first equation for t as a function of x , and then substitute this expression for t into the second equation:

From $x = 2t - 3$, we get $t = \frac{1}{2}(x + 3)$ where t is any real number.

Substitute it in $y = 6t - 7$, then we have

$$y = 6 \cdot \frac{1}{2}(x + 3) - 7 = 3x + 9 - 7$$

$$y = 3x + 2.$$

Picture????

Which is an increasing straight line (slope= $3 > 0$). The orientation is **upward, and goes through the point (0,2)**.

(b) Graph the parametric curve $x = 2t - 3$, $y = 6t - 7$ where t is from $t = 2$ to $t = -3$. by eliminating the parameter, and indicate the orientation on the graph.

From $x = 2t - 3$, $y = 6t - 7$, we get

$y = 3x + 2$, which gives you the graph. [Graph]

Orientation: If $t = 2$, then $(x, y) = (1, 5)$ and if $t = -3$, then $(x, y) = (-9, -25)$. It is a line segment from $(1, 5)$ to $(-9, -25)$, in the orientation from right to left, downward.

Exercise:3

Graph the parametric curve $x = 4t^2$, $y = 8t$ by eliminating the parameter, and indicate the orientation on the graph.

Solution: From $y = 8t$, we get $t = \frac{y}{8}$. Substitute it in $x = 4t^2$: $x = 4\left(\frac{y}{8}\right)^2$

Hence, the graph in rectangular coordinates is: $y^2 = 16x = 4(4)x$, i.e., $x = \frac{1}{16}y^2$.

Orientation: ??? [The very first parametric curve]

TANGENT LINES TO PARAMETRIC CURVES

[Recall Calculus-1: For the function $y = f(x)$, the slope of the tangent line to the curve of the function $y = f(x)$ at any x is given by $f'(x) = \frac{dy}{dx}$. It is same here.]

We will be concerned with curves that are given by parametric equations $x = f(t)$, $y = g(t)$ in which $f(t)$ and $g(t)$ have continuous first derivatives with respect to t . It can be proved that if $\frac{dx}{dt} \neq 0$, then y is a differentiable function of x , in which case the chain rule implies that

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \dots \dots \dots (1)$$

This formula makes it possible to find $\frac{dy}{dx}$ directly from the parametric equations without eliminating the parameter.

Exercise: 4

Find the slope of the tangent line to the unit circle $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$, at the point where $t = \frac{\pi}{6}$.

Solution: The slope of the tangent line is given by

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \dots \dots \dots (1)$$

Given, $x = \cos t$, $y = \sin t$

We have, $\frac{dy}{dt} = \cos t$, $\frac{dx}{dt} = -\sin t$.

So, from equation (1): $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\cos t}{-\sin t}$

At $t = \frac{\pi}{6}$, $\left[\frac{dy}{dx}\right]_{t=\frac{\pi}{6}} = \frac{\cos \frac{\pi}{6}}{-\sin \frac{\pi}{6}} = \frac{\frac{\sqrt{3}}{2}}{-\frac{1}{2}} = -\sqrt{3}$.

Hence, the slope of the tangent line is $\frac{dy}{dx} = -\sqrt{3}$.

Notes: We know that the slope tangent line to the parametric curve $x = x(t)$, $y = y(t)$ is given by

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \dots \dots \dots (1)$$

[Note: $\frac{0}{\text{non-zero}} = 0$, $\frac{\text{non-zero}}{0} = \text{undefined}$, $\frac{0}{0} = \text{indeterminant}$]

Note: A horizontal line has slope = 0, and a vertical line has slope = undefined.

(1) The tangent line to the parametric curve $x = x(t)$, $y = y(t)$ is vertical if and only if

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \text{undefined, that is, if and only if } \left[\frac{dy}{dt} \neq 0 \text{ and } \frac{dx}{dt} = 0\right].$$

(2) The tangent line to the parametric curve $x = x(t)$, $y = y(t)$ is horizontal if and only if

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = 0, \text{ that is, if and only if } \left[\frac{dy}{dt} = 0 \text{ and } \frac{dx}{dt} \neq 0\right].$$

Exercise: 5[Please solve this later]

Consider the parametric curve

$$x = (1 - \cos t) \cos t, \quad y = (1 - \cos t) \sin t, \quad 0 \leq t \leq 2\pi.$$

- Find t –values where the tangent line is horizontal.
- Find t –values where the tangent line is vertical.

Exercise: 6

Consider the parametric curve

$$x = t - 3\sin t, \quad y = 4 - 3\cos t, \quad 0 \leq t \leq 10.$$

- 1) Find t –values where the tangent line is horizontal.
- 2) Find t –values where the tangent line is vertical.

Solution: Given,

$$x = t - 3\sin t, \quad y = 4 - 3\cos t, \quad 0 \leq t \leq 10.$$

$$\frac{dx}{dt} = 1 - 3\cos t \quad \text{and} \quad \frac{dy}{dt} = 3 \sin t$$

- 1) Find t –values where the tangent line is horizontal.

Solution:

The tangent line to the parametric curve $x = x(t), y = y(t)$ is horizontal if and only if

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = 0, \text{ that is, if and only if } \left[\frac{dy}{dt} = 0 \text{ and } \frac{dx}{dt} \neq 0 \right].$$

Setting $\frac{dy}{dt} = 0$ yields the equation $3 \sin t = 0$, or, more simply, $\sin t = 0$.

This equation has four solutions in the time interval $0 \leq t \leq 10$:

$$t = 0, \quad t = \pi, \quad t = 2\pi, \quad t = 3\pi$$

Since $\frac{dx}{dt} = 1 - 3\cos t \neq 0$ for these values of t , hence the t –values are $t = 0, t = \pi, t = 2\pi,$

$t = 3\pi$ where the tangent line is horizontal.

- 2) Solution:

$$\frac{dx}{dt} = 1 - 3\cos t \quad \text{and} \quad \frac{dy}{dt} = 3 \sin t$$

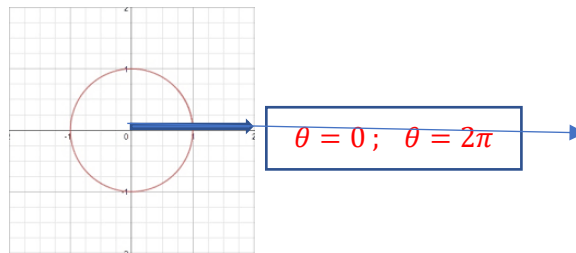
The tangent line to the parametric curve $x = x(t), y = y(t)$ is vertical if and only if

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \text{undefined, that is, if and only if } \left[\frac{dy}{dt} \neq 0 \text{ and } \frac{dx}{dt} = 0 \right]$$

Setting $\frac{dx}{dt} = 0$ yields the equation $1 - 3\cos t = 0$ or $\cos t = \frac{1}{3}$.

This equation has three solutions in the time interval $0 \leq t \leq 10$:

$$t = \cos^{-1}\left(\frac{1}{3}\right), \quad t = 2\pi - \cos^{-1}\left(\frac{1}{3}\right), \quad t = 2\pi + \cos^{-1}\left(\frac{1}{3}\right)$$



Periodic Function: $f(x) = f(p + x) = \dots = f(np + x)$; for integer p .

DONE

Note: Note that the value of $\cos^{-1}\left(\frac{1}{3}\right)$ is approximately 1.23

Length of a parametric curve:

Definition: If no segment of the parametric curve

$$x = x(t), \quad y = y(t), \quad a \leq t \leq b$$

traced more than once as t increases from a to b , then the length of the curve is defined by

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Step: Given a parametric curve $x = x(t)$, $y = y(t)$, $a \leq t \leq b$

Step 1: Find $\frac{dx}{dt}$ and $\frac{dy}{dt}$

Step 2: Find $\left(\frac{dx}{dt}\right)^2$ and $\left(\frac{dy}{dt}\right)^2$

Step 3: Find $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$

Step 4: Find $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$. Here we should try, if possible, to write $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$ as a perfect square so that we can cancel the square root.

Step 5: Evaluate the integral to find the length L:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Exercise : 7

Find the length of the curve $x = 3 \cos(2\theta)$, $y = 3 \sin(2\theta)$ for $\frac{\pi}{4} \leq \theta \leq \frac{\pi}{2}$.

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{(-6 \sin t)^2 + (6 \cos t)^2} = 6$$

Area of the Surface of Revolution by Revolving a Parametric Curve:

Definition:

- A. **Let $x = x(t)$, $y = y(t)$, $a \leq t \leq b$ be a smooth parametric curve. Then the area of the surface of revolution formed by revolving the curve about the x -axis is defined by**

$$\text{Surface Area} = \int_a^b 2\pi y(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

- B. **Let $x = x(t)$, $y = y(t)$, $a \leq t \leq b$ be a smooth parametric curve. Then the area of the surface of revolution formed by revolving the curve about the y -axis is defined by**

$$\text{Area} = \int_a^b 2\pi x(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Steps: Given a parametric curve $x = x(t)$, $y = y(t)$, $a \leq t \leq b$

Step 1: Find $\frac{dx}{dt}$ and $\frac{dy}{dt}$

Step 2: Find $\left(\frac{dx}{dt}\right)^2$ and $\left(\frac{dy}{dt}\right)^2$

Step 3: Find $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$

Step 4: Find $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$. Here we should try, if possible, to write $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$ as a perfect square so that we can cancel the square root.

Step 5: Evaluate the integral to find the area of the surface of revolution.

Exercise: 8

Find the area of the surface generated by revolving the curve $x = e^t \cos t$, $y = e^t \sin t$, ($0 \leq t \leq \pi/2$) about the x -axis.

Solution:

$$\text{Surface Area} = \int_a^b 2\pi y(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \dots \dots (1)$$

Given curve $x = e^t \cos t$, $y = e^t \sin t$, ($0 \leq t \leq \pi/2$)

$$\frac{dx}{dt} = e^t (-\sin t) + \cos t e^t = e^t (\cos t - \sin t)$$

$$\frac{dy}{dt} = e^t (\cos t) + \sin t e^t = e^t (\cos t + \sin t)$$

$$\begin{aligned} \text{So, } \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= e^{2t}[(\cos t - \sin t)^2 + (\cos t + \sin t)^2] \\ &= e^{2t}[\cos^2 t - 2 \cos t \sin t + \sin^2 t + \cos^2 t + 2 \cos t \sin t + \sin^2 t] \\ &= e^{2t}[2(\cos^2 t + \sin^2 t)] = 2e^{2t} \end{aligned}$$

$$\text{Therefore, } \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{2e^{2t}} = \sqrt{2} e^t$$

The area of the surface of revolution formed by revolving the curve about the x -axis is defined by

$$Area = \int_a^b 2\pi y(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^{\frac{\pi}{2}} 2\pi \cdot e^t \sin t \cdot \sqrt{2} e^t dt$$

$$Area = 2\sqrt{2}\pi \int_0^{\frac{\pi}{2}} e^{2t} \sin t dt \dots \dots \dots (2)$$

Here, Set $I = \int e^{2t} \sin t dt$; $u = e^{2t}$, $dv = \sin t dt$. So, $du = 2e^{2t} dt$, $v = -\cos t$

$$= \int u dv = uv - \int v du = e^{2t}(-\cos t) - \int (-\cos t) 2e^{2t} dt$$

$$= -e^{2t} \cos t + 2 \int e^{2t} \cos t dt ; u = e^{2t}, dv = \cos t dt. \text{ So, } du = 2e^{2t} dt, v = \sin t$$

$$= -e^{2t} \cos t + 2[e^{2t} \sin t - \int \sin t \cdot 2e^{2t} dt]$$

$$= -e^{2t} \cos t + 2e^{2t} \sin t - 4 \int e^{2t} \sin t dt$$

$$\Rightarrow I = e^{2t}[\sin t - \cos t] - 4I$$

$$\Rightarrow 5I = e^{2t}[\sin t - \cos t]$$

$$\Rightarrow I = \frac{e^{2t}(\sin t - \cos t)}{5} + C$$

$$\int e^{2t} \sin t dt = \frac{e^{2t}}{5} [\sin t - \cos t] + C$$

From (2):

$$Area = 2\sqrt{2}\pi \int_0^{\frac{\pi}{2}} e^{2t} \sin t dt = 2\sqrt{2}\pi \left[\frac{e^{2t}}{5} (\sin t - \cos t) \right]_0^{\frac{\pi}{2}} = \frac{2\sqrt{2}}{5} [e^{2t}(\sin t - \cos t)]_0^{\frac{\pi}{2}}$$

$$= \frac{2\sqrt{2}}{5} [e^{\frac{2\pi}{2}} (\sin \frac{\pi}{2} - \cos \frac{\pi}{2}) - e^0 (\sin 0 - \cos 0)] = \frac{2\sqrt{2}}{5} [e^{\frac{2\pi}{2}} (1 - 0) - e^0 (0 - 1)]$$

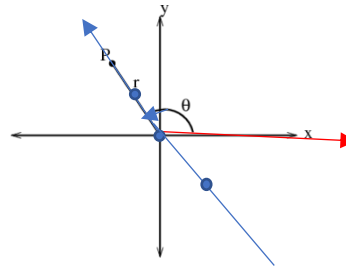
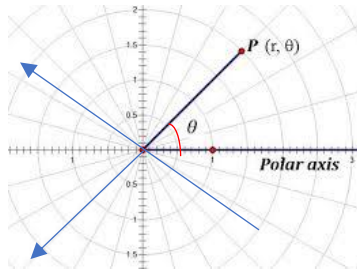
$$= \frac{2\sqrt{2}\pi}{5} \left[e^{\frac{2\pi}{2}} + e^0 \right]$$

$$Area = \frac{2\sqrt{2}\pi}{5} [e^{\pi} + 1] \text{ unit}^2$$

10.2: Polar Coordinates (r, θ)

Polar Coordinates is a **2-dimensional** coordinate space that is equivalent to Rectangular. The coordinates denoted by **(r, θ)** .

There is **only one axis** in polar coordinates given by the positive x -axis. **The point origin is called the pole.**



The above figures illustrate the polar coordinates (r, θ) of the point P .

The polar coordinates of the point P is denoted by (r, θ) , where

r : the distance from the origin to the point P , and

θ : the angle between the positive x -axis and the line segment from the origin to P .

Relation Between Rectangular (x, y) and Polar (r, θ) coordinates:

(1) Given polar coordinates (r, θ) , to find the rectangular coordinates (x, y) ,

$$\text{Set } x = r \cos \theta, \quad y = r \sin \theta$$

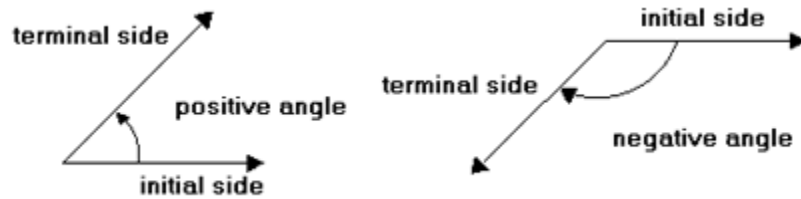
(2) Given the rectangular coordinates (x, y) , to find the polar coordinates (r, θ) , set

$$r = \sqrt{x^2 + y^2} \quad \text{or} \quad r^2 = x^2 + y^2, \quad \text{and} \quad \theta = \begin{cases} \tan^{-1}(\frac{y}{x}) & \text{if } (x, y) \text{ is in Quadrant I or IV} \\ \pi + \tan^{-1}(\frac{y}{x}) & \text{if } (x, y) \text{ is in Quadrant II or III} \end{cases}$$

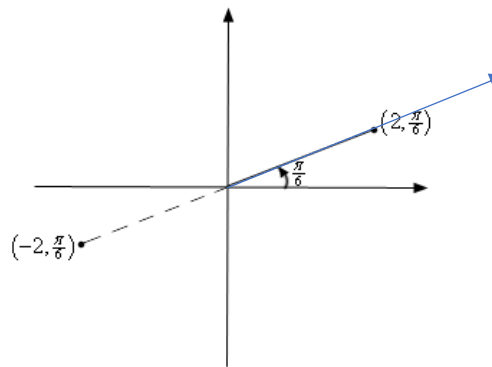
Note that the **range** of the function $\theta = \tan^{-1}(\frac{y}{x})$ is $(-\frac{\pi}{2}, \frac{\pi}{2})$

Note:

- (1) **The angle θ must be in radian.** θ is positive in anti-clockwise direction, and θ is negative in clockwise direction. The initial side of the angle θ is the polar axis.



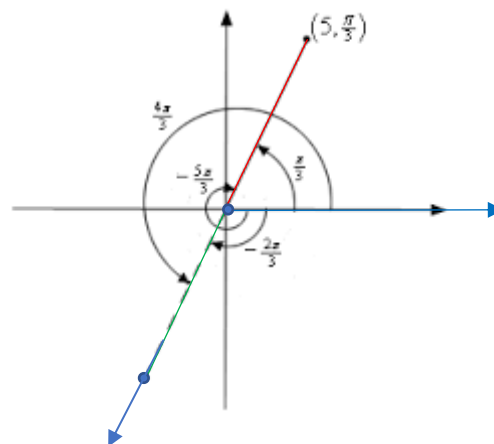
- (2) r is always non-negative, that is, $r \geq 0$. $r = 0$ only at the pole (origin). So, $-r \leq 0$. To find the coordinates $(-r, \theta)$, we must go to the backward direction. Note that the polar coordinate is a vector coordinate, and θ gives the direction. Backward direction means, beyond the point pole (origin) in the direction of θ . That is



Understanding Angles:

Notes:

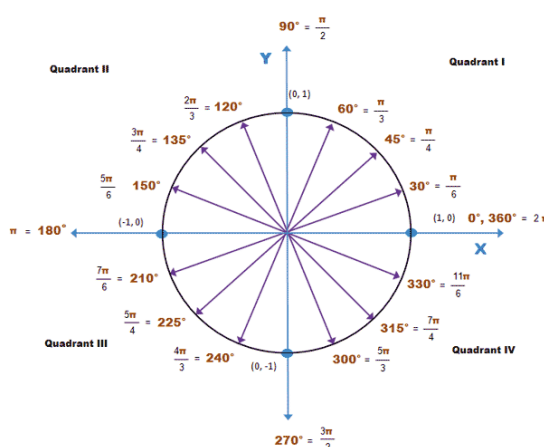
- (1) Presentation of a point in polar coordinates is not unique.



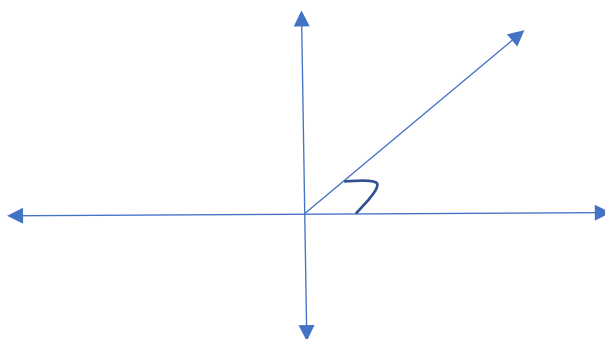
$$\left(5, \frac{\pi}{3}\right) = \left(5, 2n\pi + \frac{\pi}{3}\right) = \left(-5, \frac{4\pi}{3}\right) = \left(-5, -\frac{2\pi}{3}\right) = \left(5, -\frac{5\pi}{3}\right); \text{ where } n \text{ is a positive integer.}$$

(2) The angle θ must be in radian. The measurement of θ always starts from the polar axis (positive x -axis), and this line is called the initial side of the angle. And this line is fixed. Other side of the angle, that gives the distance of the point from the origin, is called the terminal side of the angle. Terminal side starts from the pole in outward direction. If you go beyond the pole, then the coordinate of points on the extended part of the terminal side is of the form $(-r, \theta)$, where $r > 0$. Note that $r = 0$ only at the pole.

(3) Relation between degree and radian:



$$1^\circ = \frac{\pi}{180} \text{ rad}$$



Exercise:1

(a) Find the rectangular form of the point $\left(-2, \frac{\pi}{3}\right)$.

Solution: Given polar form is $(r, \theta) = \left(-2, \frac{\pi}{3}\right)$. To find (x, y) , set

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$\text{Now, } x = r \cos \theta = (-2) \cos \frac{\pi}{3} = -2 \left(\frac{1}{2}\right) = -1$$

$$y = r \sin \theta = (-2) \sin \frac{\pi}{3} = -2 \left(\frac{\sqrt{3}}{2} \right) = -\sqrt{3}$$

Hence the rectangular form of the given point is $(x, y) = (-1, -\sqrt{3})$.

[Do not write the coordinates in decimal form]

[Given the rectangular coordinates (x, y) , to find the polar coordinates (r, θ) , set

$$r = \sqrt{x^2 + y^2} \quad \text{or} \quad r^2 = x^2 + y^2, \quad \text{and} \quad \theta = \begin{cases} \tan^{-1} \frac{y}{x} & \text{if } (x, y) \text{ is in Quadrant I or IV} \\ \pi + \tan^{-1} \frac{y}{x} & \text{if } (x, y) \text{ is in Quadrant II or III} \end{cases}$$

(b) Find the polar form of the point $(-2, 2\sqrt{3})$.

Solution: Given the rectangular point $(x, y) = (-2, 2\sqrt{3})$. This point is in Quadrant II.

So, set $r = \sqrt{x^2 + y^2}$ and $\theta = \pi + \tan^{-1} \left(\frac{y}{x} \right)$

$$\text{Now, } r = \sqrt{x^2 + y^2} = \sqrt{(-2)^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = 4.$$

And,

$$\theta = \pi + \tan^{-1} \left(\frac{y}{x} \right) = \pi + \tan^{-1} \left(\frac{2\sqrt{3}}{-2} \right) = \pi + \tan^{-1}(-\sqrt{3}) = \pi - \tan^{-1}(\sqrt{3}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

Hence the polar form of the point is $\left(4, \frac{2\pi}{3} \right)$.

Note: $y = \tan^{-1}x$ is an odd function, and hence we get $\tan^{-1}(-x) = -\tan^{-1}x$;

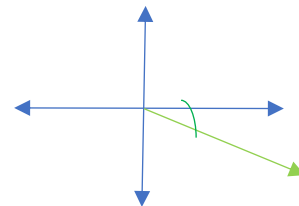
Odd Function: $f(-x) = -f(x)$

(c) Find the polar form of the point $(3, -\sqrt{3})$. [Please do it!!]

Solution: Given the rectangular point $(x, y) = (3, -\sqrt{3})$. This point is in Quadrant IV.

So, set $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \frac{y}{x}$

$$\text{Ans: } (r, \theta) = \left(2\sqrt{3}, \frac{11\pi}{6} \right)$$



(d) Find the polar form of the point $(3, -3)$. [Please do it!!]

Solution: Given the rectangular point $(x, y) = (3, -3)$. This point is in Quadrant IV.

So, set $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \frac{y}{x}$

Ans: $(r, \theta) = \left(3\sqrt{2}, \frac{7\pi}{4}\right)$

Summery:

- (1) Definition of polar coordinates
- (2) Relation between the polar coordinates and the rectangular coordinates
- (3) Type of the coordinates of a point which is on the extended part of the terminal side of an angle θ .
- (4) We did one exercise. **[Exercise-1]**

Converting Equations of Polar Coordinates

Exercise: 2

Find the equations of the following curves in polar coordinates:

- (1) $x^2 + y^2 = 4$
- (2) $x^2 + 2x + y^2 = 8$ (Complete!)
- (3) $y = x$ (please do it!)
- (4) $y = \sqrt{3}x$
- (5) $x^2 + 2x + y^2 = 0$



Solution: (1) We know that $x^2 + y^2 = r^2$

$$x^2 + y^2 = 4 \Rightarrow r^2 = 4 \quad \therefore \quad r = 2$$

Hence, $r = 2$ gives us the circle with center at the origin $(0, 0)$ and with radius 2.

Note: Let the constant $k > 0$. Then $r = k$ gives us the circle with center at the origin $(0, 0)$ and with radius k in polar coordinates.

Solution (5): Given $x^2 + 2x + y^2 = 0 \dots (1)$

[Note that $x^2 + y^2 + 2x = 0$ can be written as $(x + 1)^2 + y^2 = 1^2$ which is a circle of radius 1 with center at the point $(-1, 0)$.]

Now, we know that $x = r \cos \theta$, $y = r \sin \theta \Rightarrow x^2 + y^2 = r^2$.

Then from (1):

$$\begin{aligned} x^2 + y^2 + 2x &= 0 \\ \Rightarrow r^2 + 2r \cos \theta &= 0 \\ \Rightarrow r^2 &= -2r \cos \theta \\ \therefore r &= -2 \cos \theta. \end{aligned}$$

Note: $r = a \cos \theta$ is a circle with center on the x -axis at the point $\left(\frac{a}{2}, 0\right)$ with radius $\left|\frac{a}{2}\right|$.

Hence, $r = -2 \cos \theta$ gives us a circle with center at the point $\left(-\frac{2}{2}, 0\right) = (-1, 0)$ and radius $\left|-\frac{2}{2}\right| = 1$.

Note: $r = a \sin \theta$ is a circle with center on the y -axis at the point $\left(0, \frac{a}{2}\right)$ with radius $\left|\frac{a}{2}\right|$.

Note: $r = k$, k is a positive constant : the circle with center at the origin $(0, 0)$ and with radius k .

Solution: (2) $x^2 + 2x + y^2 = 8$

$$\Rightarrow x^2 + y^2 + 2x = 8$$

$$\Rightarrow r^2 + 2r \cos \theta = 8$$

$$\Rightarrow r^2 + 2r \cos \theta - 8 = 0$$

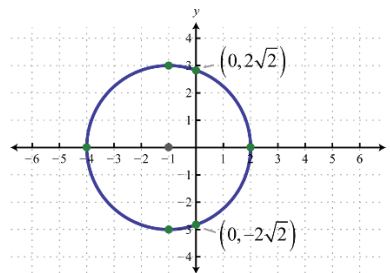
Complete : Solve for r .

Note: A Polar curve is usually denoted by $r = f(\theta)$

[Home task:

Consider $r^2 + 2r \cos \theta - 8 = 0$ as $x^2 + 2x \cos \theta - 8 = 0$ and use the formula for the Quadratic equation to solve for r .

Formula: For $ax^2 + bx + c = 0$, we get $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.



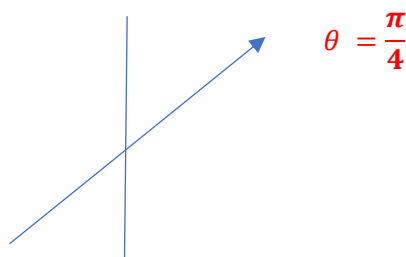
$$x^2 + 2x + y^2 = 8 \text{ Or } (x + 1)^2 + y^2 = 3^2$$

Solution: (3) $y = x \Rightarrow r \sin \theta = r \cos \theta$

$$\Rightarrow \frac{r \sin \theta}{r \cos \theta} = 1 \Rightarrow \tan \theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

Here, $\theta = \frac{\pi}{4}$ is the equation of the line $y = x$. In polar form $\theta = \frac{\pi}{4}$ is a ray, a line with one direction. The direction is given by the terminal side of θ .





Note: If $\theta = \theta_0$, θ_0 is a constant: A ray from the pole to infinity in outward direction.

Solution: (4)

Given $y = \sqrt{3}x \dots \dots (1)$

We know that $x = r \cos \theta$ and $y = r \sin \theta$

$$y = \sqrt{3}x \Rightarrow r \sin \theta = \sqrt{3} r \cos \theta$$

$$\Rightarrow r \sin \theta - \sqrt{3} r \cos \theta = 0$$

$$\Rightarrow r(\sin \theta - \sqrt{3} \cos \theta) = 0$$

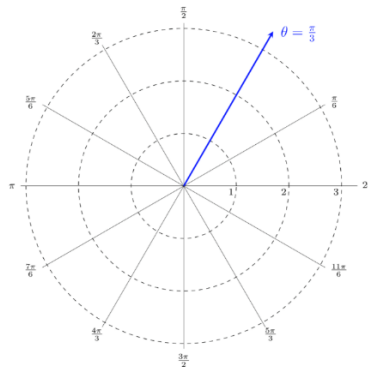
$$\Rightarrow r = 0 \text{ or } (\sin \theta - \sqrt{3} \cos \theta) = 0$$

$$\Rightarrow r = 0 \text{ or } \sin \theta = \sqrt{3} \cos \theta$$

$$\Rightarrow r = 0, \frac{\sin \theta}{\cos \theta} = \sqrt{3} \Rightarrow r = 0, \tan \theta = \sqrt{3}$$

$$\therefore \theta = \frac{\pi}{3}$$

So, the polar form of the equation $y = \sqrt{3}x$ is $\theta = \frac{\pi}{3}$; which is a Ray.



Definition: A ray is a one way directed line from the pole to infinity in outward orientation given by the equation $\theta = \theta_0$, θ_0 is a constant. Note that since a ray goes through the pole, hence $r = 0$ must be set to find the equation of a ray.

Polar Function: $r = f(\theta)$

Exercise: 3

Find the equations of the following curves in rectangular coordinates:

A) $r = 3$

B) $\theta = \frac{\pi}{4}$

C) $r = \sin \theta$ [please do it]

D) $r = 3 \cos \theta$

E) $\theta = \frac{5\pi}{4}$ (homework)

F) $r = -4 \cos \theta$

G) $r = -5 \sin \theta$

H) $\theta = \frac{3\pi}{4}$ (homework)

Solutions:

Part (A): Given $r = 3 \Rightarrow \sqrt{x^2 + y^2} = 3 \Rightarrow x^2 + y^2 = 3^2$, a circle with center at the origin and with radius 3.

Part (B) Given $\theta = \frac{\pi}{4} \Rightarrow \tan^{-1}\left(\frac{y}{x}\right) = \frac{\pi}{4} \Rightarrow \frac{y}{x} = \tan\left(\frac{\pi}{4}\right) \Rightarrow \frac{y}{x} = 1 \quad \therefore y = x$.

Part (E) : Given $\theta = \frac{5\pi}{4} \Rightarrow \pi + \tan^{-1}\left(\frac{y}{x}\right) = \frac{5\pi}{4}$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x}\right) = \frac{5\pi}{4} - \pi$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{x}\right) = \frac{\pi}{4}$$

$$\Rightarrow \frac{y}{x} = 1 \text{ that is } y = x.$$

Part (D) Given $r = 3 \cos \theta \dots \dots \dots (1)$

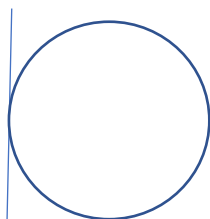
Since $x = r \cos \theta$ and $y = r \sin \theta$, then we get $x^2 + y^2 = r^2$ and $\cos \theta = \frac{x}{r}$. From (1):

$$\Rightarrow r = 3 \frac{x}{r} \Rightarrow r^2 = 3x$$

$$\Rightarrow x^2 + y^2 = 3x \Rightarrow (x^2 - 3x) + y^2 = 0 \Rightarrow \left(x^2 - 2 \cdot x \cdot \frac{3}{2} + \left(\frac{3}{2}\right)^2\right) + y^2 = \left(\frac{3}{2}\right)^2$$

$$\therefore \left(x - \frac{3}{2}\right)^2 + (y - 0)^2 = \left(\frac{3}{2}\right)^2.$$

$\left(x - \frac{3}{2}\right)^2 + (y - 0)^2 = \left(\frac{3}{2}\right)^2$ is the circle with center at the point $\left(\frac{3}{2}, 0\right)$ and radius $\frac{3}{2}$.





Note: $(x - h)^2 + (y - k)^2 = r^2$ is a circle with center at the point (h, k) and radius r .

Note: $r = \text{constant}$, $r = a \sin \theta$, $r = b \cos \theta$ are equations of circles, where a, b are constants.

$r = a \sin \theta$ is a circle of radius $\frac{|a|}{2}$ with center on the y -axis which is $(0, \frac{a}{2})$.

G) Given equation $r = -5 \sin \theta$ (1)

We know that $x = r \cos \theta$, $y = r \sin \theta \Rightarrow x^2 + y^2 = r^2$ and $\sin \theta = \frac{y}{r}$. Now, from equation (1):

$$r = -5 \sin \theta$$

$$\Rightarrow r = -5 \left(\frac{y}{r} \right)$$

$$\Rightarrow r^2 = -5y$$

$$\Rightarrow x^2 + y^2 + 5y = 0 \Rightarrow x^2 + (y^2 + 5y) = 0$$

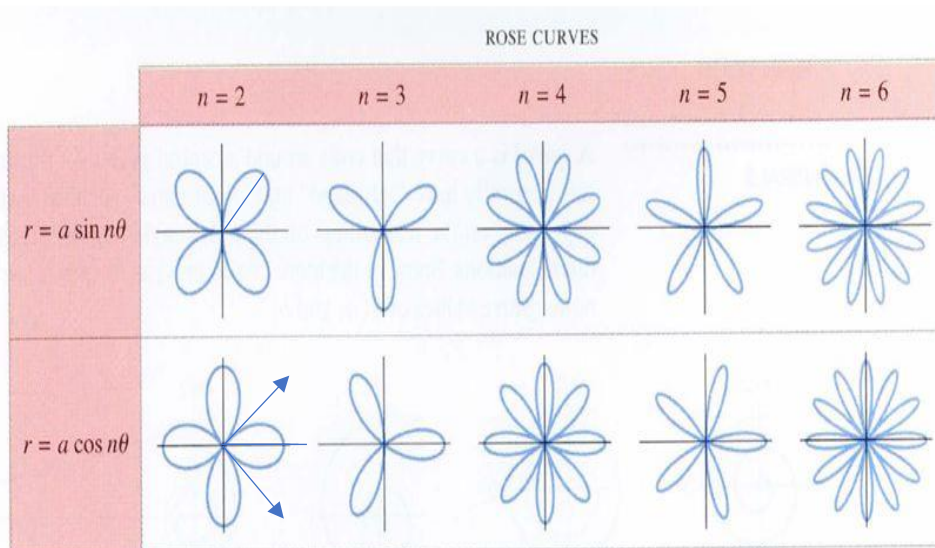
$$\Rightarrow x^2 + \left(y^2 + 2 \cdot y \cdot \frac{5}{2} + \left(\frac{5}{2} \right)^2 \right) = \left(\frac{5}{2} \right)^2$$

$$\therefore (x - 0)^2 + \left(y + \frac{5}{2} \right)^2 = \left(\frac{5}{2} \right)^2$$

Hence, the equation is a circle with center at $(0, -\frac{5}{2})$ and radius $\frac{5}{2}$.

Families of Rose Curves

- Equations of the following form are called rose curves. Notice when n is odd it is the number of rose petals. When n is even, there are $2n$ rose petals.



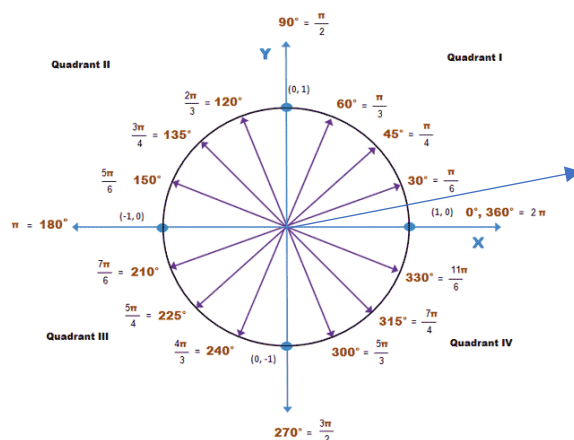
Equations: $r = a \sin(n\theta)$, $r = a \cos(n\theta)$; $n > 1$ integer, $a > 0$.

Notes:

- If n is odd, then there are n number of petals in the rose. If n is even, then there are $2n$ number of petals in the rose.
- Here $r = a \sin(n\theta)$, $r = a \cos(n\theta)$; $n > 1$, $a > 0$; a is the length of each petal.
- In the rose $r = a \cos(n\theta)$, there is one petal symmetric with respect to the polar axis (positive x -axis)
- We divide the xy -plane in $2n$ number of sectors.
- If n is odd, then there are n number of petals in the rose where petals will be placed in the sectors alternately.

(6) If n is even, then there are $2n$ number of petals in the rose where petals will be placed in each sector.

(6) Each petal is symmetric with respect to it's axis which is the bisector of the sector.



Exercise: 4/a

Graph the rose $r = 2 \sin(4\theta)$.

Exercise: 4/b

Graph the rose $r = 2 \sin(3\theta)$.

Solution: Given $r = 2 \sin(3\theta)$, $a = 2$, $n = 3$.

There are 3 petals.

Divide xy -plane in $2(3) = 6$ sectors with rays.

[Note: To find Rays, set $r = 0$.]

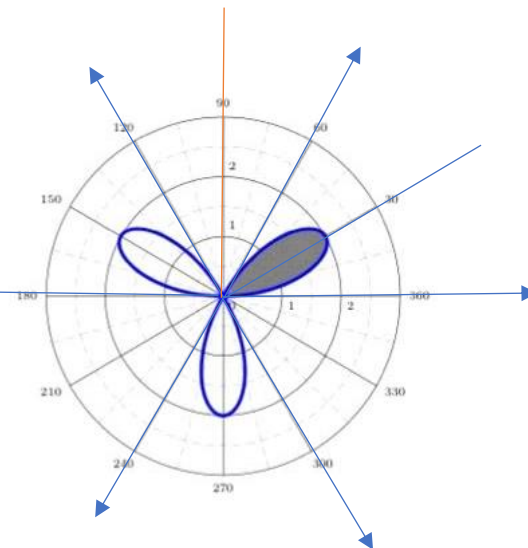
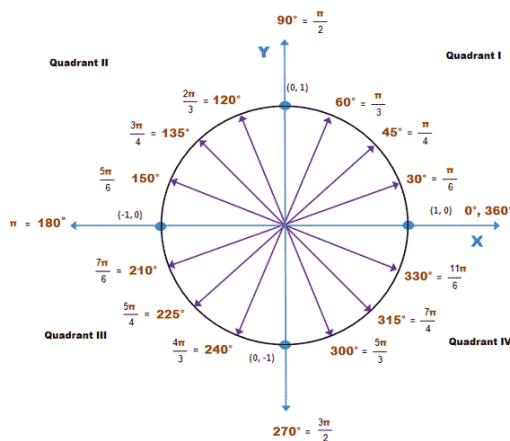
To find equations of rays, set $r = 0 \Rightarrow 2 \sin(3\theta) = 0 \Rightarrow \sin(3\theta) = 0$.

$\Rightarrow \sin(3\theta) = 0$; Note that $\sin(k\pi) = 0$ for some integer k .

$\Rightarrow 3\theta = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi$

$\therefore \theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}$ in $[0, 2\pi]$.

$$\therefore \theta = 0, \quad \frac{\pi}{3}, \quad \frac{2\pi}{3}, \quad \pi, \quad \frac{4\pi}{3}, \quad \frac{5\pi}{3}$$



Bisector of the first sector (between the rays $\theta = 0$ and $\theta = \frac{\pi}{3}$) is $\theta = \frac{\pi}{6}$: $r = 2 \sin(3\theta)$

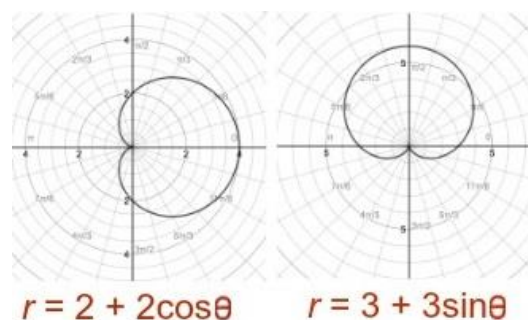
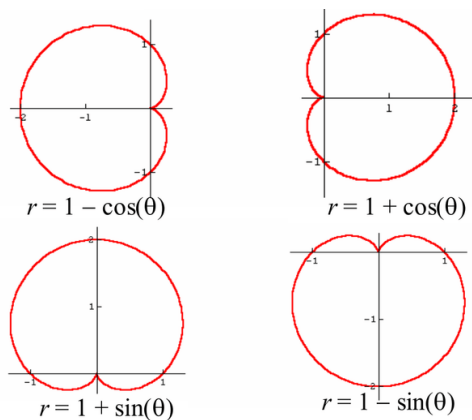
Here $r = 2 \sin(3 \cdot \frac{\pi}{6}) = 2 \sin \frac{\pi}{2} = 2$. Since $r = 2 > 0$, so this sector contains a petal. The point (vertex) on the petal (or bisector) is $(2, \frac{\pi}{6})$

Bisector of the second sector (between the rays $\theta = \frac{\pi}{3}$ and $\theta = \frac{2\pi}{3}$) is $\theta = \frac{\pi}{2}$: $r = 2 \sin(3\theta)$

$$\text{Here } r = 2 \sin(3 \cdot \frac{\pi}{2}) = 2 \sin(\frac{3\pi}{2}) = 2(-1) = -2.$$

Since $r = -2 < 0$, so this sector contains no petal. Point on the petal is $(-2, \frac{\pi}{2})$ which is in the sector opposite to it.

Families of Cardioid Curves $r = a \pm a \sin \theta = a(1 \pm \sin \theta)$, $r = a \pm a \cos \theta = a(1 \pm \cos \theta)$



$r = a \pm a \sin \theta$ symmetric with respect to y -axis.

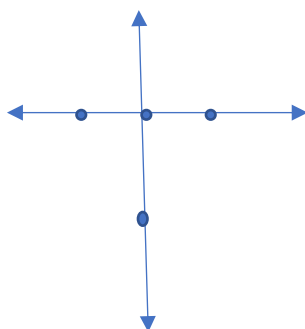
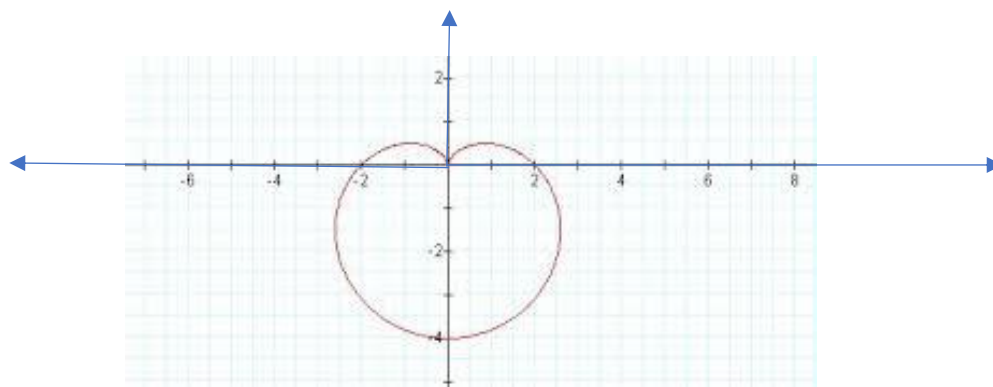
$r = a \pm a \cos \theta$ symmetric with respect to x -axis.

Exercise: 5

Graph the Cardioid $r = 2 - 2 \sin \theta$.

Solution: Given $r = 2 - 2 \sin \theta$.

θ	r	(r, θ)
0	2	$(2, 0)$
$\frac{\pi}{2}$	0	$(0, \frac{\pi}{2})$
π	2	$(2, \pi)$
$\frac{3\pi}{2}$	4	$(4, \frac{3\pi}{2})$



10.3: Polar Functions: Tangent to the Curve, Length, Area

1. Finding slopes of tangent lines
2. Finding points where the tangent line is either vertical or horizontal
3. Finding length of a polar curve
4. Finding the area of a simple polar region

Recall: (x, y) : $x = f(y)$ OR $y = f(x)$. Note that $r = 1$ is a circle. Hence, $\theta = f(r)$ not valid.

Polar Functions:

A polar function (polar curve) is given by (r, θ) : $r = f(\theta)$ in which r is a function of θ .

Parametric Form:

Given a polar curve $r = f(\theta)$.

This can be expressed parametrically in terms of the parameter θ by substituting $f(\theta)$ for r in the equations $x = r \cos \theta$ and $y = r \sin \theta$.

That is, parametric equations are $x = f(\theta) \cos \theta$, $y = f(\theta) \sin \theta$; where θ is the parameter.

TANGENT LINES TO POLAR CURVES

[Recall Calculus-1: For the function $y = f(x)$, the slope of the tangent line to the curve of the function $y = f(x)$ at any x is given by $f'(x) = \frac{dy}{dx}$.]

Polar curves are given in parametric form by the equations $x = f(\theta) \cos \theta$, $y = f(\theta) \sin \theta$; where θ is the parameter and $f(\theta)$ has continuous first derivatives with respect to θ .

It can be proved that if $\frac{dx}{d\theta} \neq 0$, then y is a differentiable function of x , in which case the chain rule implies that

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \dots \dots \dots (1)$$

This formula makes it possible to find $\frac{dy}{dx}$ directly from the parametric equations without eliminating the parameter.

Exercise: 1

Find the slope of the tangent line to the circle $r = \cos\theta$, $0 \leq \theta \leq 2\pi$, at the point where $\theta = \frac{\pi}{4}$.

Solution: The slope of the tangent line is given by

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \dots \dots \dots (1)$$

Given Polar Curve: $r = \cos\theta$

The parametric equations of the circle is given by

$$x = r \cos\theta = \cos\theta \cos\theta = \cos^2\theta$$

$$y = r \sin\theta = \cos\theta \sin\theta = \frac{1}{2} \sin(2\theta)$$

[Chain Rule: $\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$. For $\frac{d}{d\theta} (\cos\theta)^2$, $f(\theta) = \theta^2$, $g(\theta) = \cos\theta$]

Since,

$$x = \cos^2\theta$$

$$\text{We have, } \frac{dx}{d\theta} = \frac{d}{d\theta} (\cos^2\theta) = \frac{d}{d\theta} (\cos\theta)^2 = 2 \cos\theta (-\sin\theta) = -\sin(2\theta) ,$$

And $y = \frac{1}{2} \sin(2\theta)$, so we get

$$\frac{dy}{d\theta} = \frac{d}{d\theta} \left(\frac{1}{2} \sin(2\theta) \right) = \frac{1}{2} \cos(2\theta) \cdot 2 = \cos(2\theta)$$

So, from equation (1):

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\cos(2\theta)}{-\sin(2\theta)}$$

At $\theta = \frac{\pi}{4}$,

$$\left[\frac{dy}{dx} \right]_{\theta=\frac{\pi}{4}} = \frac{\cos \frac{\pi}{2}}{-\sin \frac{\pi}{2}} = \frac{0}{-1} = 0 .$$

Hence, the slope of the tangent line is $\frac{dy}{dx} = 0$.

That is, the tangent line to the polar curve $r = \cos\theta$ at $\theta = \frac{\pi}{4}$ is horizontal.

Done!!!

Note: $\frac{0}{\text{non-zero}} = 0$, $\frac{\text{non-zero}}{0} = \text{undefined}$, $\frac{0}{0} = \text{indeterminant}$

Notes:

The slope of the tangent line is given by

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} \dots \dots \dots (1)$$

(3) The tangent line to the parametric curve $x = f(\theta)\cos\theta$, $y = f(\theta)\sin\theta$ is vertical if and only if

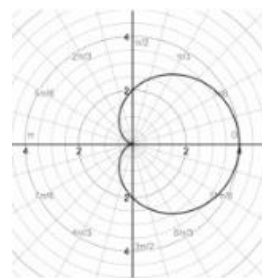
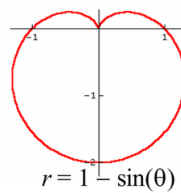
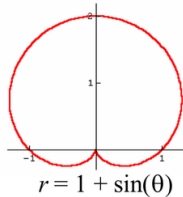
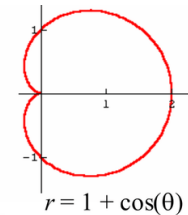
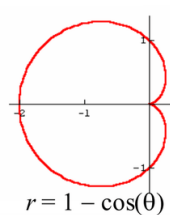
$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \text{undefined, that is, if and only if } \left[\frac{dy}{d\theta} \neq 0 \text{ and } \frac{dx}{d\theta} = 0 \right].$$

(4) The tangent line to the parametric curve $x = f(\theta)\cos\theta$, $y = f(\theta)\sin\theta$ is **horizontal** if and only if

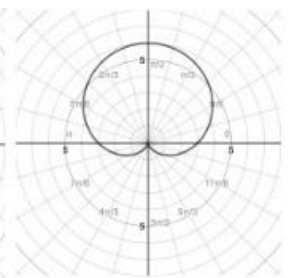
$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = 0, \text{ that is, if and only if } \left[\frac{dy}{d\theta} = 0 \text{ and } \frac{dx}{d\theta} \neq 0 \right].$$

Families of Cardioid Curves:

$$r = a(1 \pm \cos \theta), \quad r = a(1 \pm \sin \theta)$$



$$r = 2 + 2\cos\theta$$



$$r = 3 + 3\sin\theta$$

Observation:

- (1) $r = a(1 \pm \cos \theta)$ is symmetric with respect to x -axis.
- (2) $r = a(1 \pm \sin \theta)$ is symmetric with respect to y -axis.
- (3) Domain of a Cardioid is $[0, 2\pi]$

[10.1 → Exercise: 5[Please solve this later]

Consider the parametric curve

$$x = (1 - \cos t) \cos t, \quad y = (1 - \cos t) \sin t, \quad 0 \leq t \leq 2\pi.$$

- Find t –values where the tangent line is horizontal.
- Find t –values where the tangent line is vertical.]

Exercise: 2

Find the values of θ where the tangent line to the Cardioid $r = 1 - \cos \theta$ is either horizontal or vertical.

Solution: We know that

- (i) vertical tangent iff $\left[\frac{dy}{d\theta} \neq 0 \text{ and } \frac{dx}{d\theta} = 0 \right]$
- (ii) Horizontal tangent iff $\left[\frac{dy}{d\theta} = 0 \text{ and } \frac{dx}{d\theta} \neq 0 \right]$

Consider the parametric of the polar curve $r = 1 - \cos \theta$

$$x = (1 - \cos \theta) \cos \theta, \quad y = (1 - \cos \theta) \sin \theta, \quad 0 \leq \theta \leq 2\pi.$$

$$\frac{dx}{d\theta} = (1 - \cos \theta) \frac{d}{d\theta}(\cos \theta) + \cos \theta \frac{d}{d\theta}(1 - \cos \theta)$$

$$= (1 - \cos \theta) (-\sin \theta) + \cos \theta \sin \theta$$

$$= -\sin \theta + 2 \sin \theta \cos \theta$$

$$\therefore \frac{dx}{d\theta} = \sin \theta (2 \cos \theta - 1)$$

$$y = (1 - \cos \theta) \sin \theta$$

$$\frac{dy}{d\theta} = (1 - \cos \theta) \frac{d}{d\theta}(\sin \theta) + \sin \theta \frac{d}{d\theta}(1 - \cos \theta)$$

$$= (1 - \cos \theta) \cos \theta + \sin \theta \sin \theta = \cos \theta - \cos^2 \theta + \sin^2 \theta$$

$$= \cos \theta - \cos^2 \theta + 1 - \cos^2 \theta$$

$$= 1 + \cos \theta - 2 \cos^2 \theta \quad ; \quad [\text{Think: } p = \cos \theta ; 1 + p - 2p^2]$$

$$= 1 - \cos \theta + 2 \cos \theta - 2 \cos^2 \theta$$

$$= 1(1 - \cos \theta) + 2 \cos \theta (1 - \cos \theta)$$

$$\therefore \frac{dy}{d\theta} = (1 - \cos \theta)(1 + 2 \cos \theta)$$

Now, Set, $\frac{dx}{d\theta} = \sin \theta (2\cos \theta - 1) = 0$, then $\sin \theta = 0$, $2\cos \theta - 1 = 0$

$$\Rightarrow \sin \theta = 0, \quad \cos \theta = \frac{1}{2}; \quad 0 \leq \theta \leq 2\pi.$$

We know that, $\sin \theta = 0 \Rightarrow \theta = 0, \pi, 2\pi$, and $\cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}, 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

$$\frac{dx}{d\theta} = 0 : \quad \theta = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

Again, Set $\frac{dy}{d\theta} = (1 - \cos \theta)(1 + 2\cos \theta) = 0$,

$$\text{then } \cos \theta = 1, \quad \cos \theta = -\frac{1}{2} \text{ for } 0 \leq \theta \leq 2\pi.$$

$$[\text{Think: } \cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3} \left\{ \cos\left(\pi \pm \frac{\pi}{3}\right) = -\cos\left(\frac{\pi}{3}\right) \right\}]$$

Here, $\cos \theta = 1 \Rightarrow \theta = 0, 2\pi$

$$\text{and } \cos \theta = -\frac{1}{2} = -\cos \frac{\pi}{3}$$

$$\Rightarrow \cos \theta = \cos\left(\pi \pm \frac{\pi}{3}\right); \quad [-\cos \varphi = \cos(\pi \pm \varphi) \text{ where } \varphi \text{ is an acute angle.}]$$

$$\Rightarrow \theta = \pi \pm \frac{\pi}{3}, \text{ that is, } \theta = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\frac{dy}{d\theta} = 0: \quad \theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}, 2\pi$$

Now:

$$\frac{dy}{d\theta} = 0: \quad \theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}, 2\pi, \text{ and}$$

$$\frac{dx}{d\theta} = 0: \quad \theta = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

(a) The tangent line is vertical iff $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \text{undefined}$, that is, iff $\theta = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$

(b) The tangent line is horizontal iff $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = 0$, that is, iff $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$

Exercise: 3 (homework)

Consider the parametric curve $x = 3 \sin \theta$, $y = 4 \cos \theta$, $0 \leq \theta \leq \pi$.

3) Find θ -values where the tangent line is horizontal.

4) Find θ -values where the tangent line is vertical.

Solution: ???

Arc Length of a polar curve:

Definition: If no segment of the polar curve $r = f(\theta)$ is traced more than once as θ increases from α to β , and if $\frac{dr}{d\theta} = f'(\theta)$ is continuous for $\alpha \leq \theta \leq \beta$, then the arc length L from $\theta = \alpha$ to $\theta = \beta$ is

$$L = \int_{\alpha}^{\beta} \sqrt{(f(\theta))^2 + (f'(\theta))^2} d\theta = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \dots \dots (2)$$

Exercise: 4 (homework)

Find the arc length of the spiral $r = e^{3\theta}$ between $\theta = 0$ and $\theta = \pi$.

Note: $\sqrt{e^{6\theta}} = e^{\frac{6\theta}{2}} = e^{3\theta}$; Property of exponent $(a^n)^m = a^{nm}$

Solution:

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \dots \dots (1)$$

Given $r = e^{3\theta}$, $\frac{dr}{d\theta} = 3e^{3\theta}$. Then, $r^2 = (e^{3\theta})^2 = e^{6\theta}$

and $\left(\frac{dr}{d\theta}\right)^2 = (3e^{3\theta})^2 = 9e^{6\theta}$, $0 \leq \theta \leq \pi$

$$L = \int_0^{\pi} \sqrt{e^{6\theta} + 9e^{6\theta}} d\theta$$

$$= \int_0^{\pi} \sqrt{10e^{6\theta}} d\theta \quad ; \quad \text{Note: } \sqrt{e^{6\theta}} = e^{\frac{6\theta}{2}} = e^{3\theta} > 0$$

$$= \int_0^{\pi} \sqrt{10} e^{3\theta} d\theta$$

$$= \frac{\sqrt{10}}{3} [e^{3\theta}]_0^{\pi} = \frac{\sqrt{10}}{3} (e^{3\pi} - 1) \text{ unit}$$

Exercise: 5/a

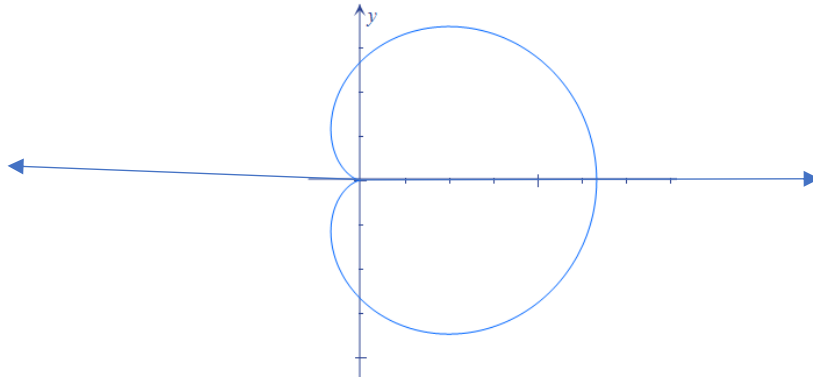
Find the total arc length of the Cardioid $r = 1 + \cos\theta$.

Solution:

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \dots \dots (1)$$

Given Cardioid $r = 1 + \cos\theta$. Hence $0 \leq \theta \leq 2\pi$.

[$r = a(1 \pm \cos\theta)$ is symmetric with respect to x -axis.]



$$\frac{dr}{d\theta} = -\sin\theta.$$

Now, from equation (1), Length of the Cardioid,

$$\begin{aligned} L &= \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \\ &= 2 \int_0^{\pi} \sqrt{(1 + \cos\theta)^2 + (-\sin\theta)^2} d\theta \\ &= 2 \int_0^{\pi} \sqrt{1 + 2\cos\theta + \cos^2\theta + \sin^2\theta} d\theta \\ &= 2 \int_0^{\pi} \sqrt{2 + 2\cos\theta} d\theta \\ &= 2 \int_0^{\pi} \sqrt{2(1 + \cos\theta)} d\theta \\ &= 2 \int_0^{\pi} \sqrt{2 \left(1 + \cos\left(2\left(\frac{\theta}{2}\right)\right)\right)} d\theta \end{aligned}$$

$$= 2 \int_0^{\pi} \sqrt{2 \cdot 2 \cos^2 \left(\frac{\theta}{2} \right)} d\theta ;$$

$$[\cos^2 x = \frac{1}{2}[1 + \cos(2x)] \Rightarrow 1 + \cos(2x) = 2 \cos^2 x. \text{ If } x = \frac{\theta}{2}, \text{ then } 2x = \theta.]$$

$$= 2 \int_0^{\pi} \sqrt{\left(2 \cos \frac{\theta}{2} \right)^2} d\theta ; \quad \sqrt{x^2} = |x| \text{ for any real number } x.$$

$$= 2 \int_0^{\pi} 2 \left| \cos \frac{\theta}{2} \right| d\theta$$

$$= 4 \int_0^{\pi} \left| \cos \left(\frac{\theta}{2} \right) \right| d\theta ; \text{ Since } 0 \leq \theta \leq \pi, \text{ hence } 0 \leq \frac{\theta}{2} \leq \frac{\pi}{2}.$$

$$= 4 \int_0^{\pi} \cos \left(\frac{\theta}{2} \right) d\theta ; \quad \cos \frac{\theta}{2} \geq 0 \text{ for } 0 \leq \frac{\theta}{2} \leq \frac{\pi}{2}.$$

$$= 4 \int_0^{\pi} \cos \frac{\theta}{2} d\theta = 4 \left[\frac{\sin \frac{\theta}{2}}{\frac{1}{2}} \right]_0^{\pi}$$

$$= 8 \left[\sin \frac{\theta}{2} \right]_0^{\pi} = 8(1 - 0) = 8 \text{ unit}$$

Exercise: 5/b

Find the length of the portion of the Cardioid $r = 1 + \cos \theta$, $\pi \leq \theta \leq 2\pi$.

$$L = 2 \int_{\pi}^{2\pi} \left| \cos \left(\frac{\theta}{2} \right) \right| d\theta ; \text{ Since } \pi \leq \theta \leq 2\pi, \text{ hence } \frac{\pi}{2} \leq \frac{\theta}{2} \leq \pi \text{ and hence } \cos \left(\frac{\theta}{2} \right) \leq 0.$$

$$L = 2 \int_{\pi}^{2\pi} -\cos \left(\frac{\theta}{2} \right) d\theta$$

Exercise: 5/c

Find the total arc length of the cardioid $r = 1 \pm \sin \theta$.

[Note that the Cardioid $r = a(1 \pm \sin \theta)$ is symmetric with respect to y - axis.]

Solution:

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \dots \dots (1)$$

$$L = 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta = 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\left(\cos\left(\frac{\theta}{2}\right) \pm \sin\left(\frac{\theta}{2}\right)\right)^2} d\theta$$

$$= 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left| \cos\left(\frac{\theta}{2}\right) \pm \sin\left(\frac{\theta}{2}\right) \right| d\theta \dots \dots (2)$$

Note: $1 \pm \sin \theta = \sin^2\left(\frac{\theta}{2}\right) + \cos^2\left(\frac{\theta}{2}\right) \pm 2 \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) = \left(\cos\left(\frac{\theta}{2}\right) \pm \sin\left(\frac{\theta}{2}\right)\right)^2$;

$[\sin(2x) = 2 \sin x \cos x]$ Note: $\cos(x) \geq \sin(x)$ for $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

Complete!

TANGENT LINES TO POLAR CURVES AT THE ORIGIN

Formula (1) reveals some useful information about the behavior of a polar curve $r = f(\theta)$ that passes through the origin. If we assume that $r = 0$ and $f'(\theta) = \frac{dr}{d\theta} \neq 0$ when $\theta = \theta_0$, then it follows from Formula (1) that the slope of the tangent line to the parametric curve

$x = f(\theta)\cos\theta$, $y = f(\theta)\sin\theta$ at $\theta = \theta_0$ is

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}]_{\theta=\theta_0}}{\frac{dx}{d\theta}]_{\theta=\theta_0}} = \frac{0 + \sin\theta_0 \frac{dr}{d\theta}}{0 + \cos\theta_0 \frac{dr}{d\theta}} = \frac{\sin\theta_0}{\cos\theta_0} = \tan\theta_0$$

However, $\tan\theta_0$ is also the slope of the line $\theta = \theta_0$, so we can conclude that this line is tangent to the curve at the origin. Thus, we have established the following result]

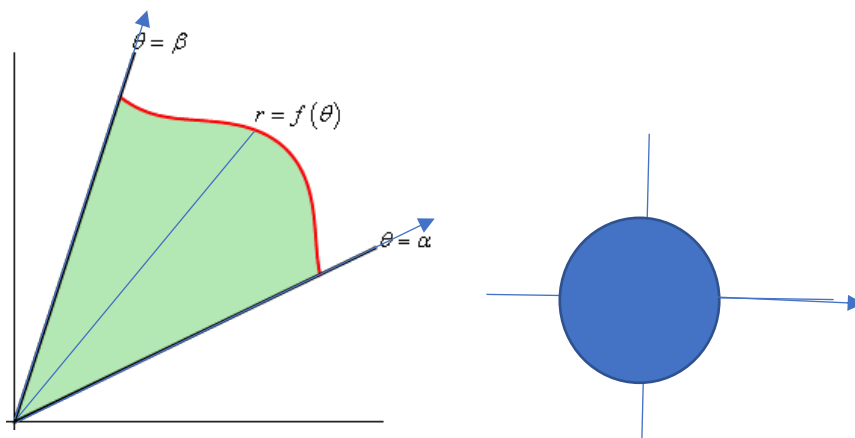
DAY-23

Theorem: 1 If the polar curve $r = f(\theta)$ passes through the origin at $\theta = \theta_0$, and if $\frac{dr}{d\theta} \neq 0$ at $\theta = \theta_0$, then the line $\theta = \theta_0$ is tangent to the curve at the origin.

Area in polar coordinates:

If α and β are angles that satisfy the condition $\alpha < \beta \leq \alpha + 2\pi$ and if $f(\theta)$ is continuous and either non-negative or non-positive for $\alpha \leq \theta \leq \beta$, then the area A of the region R enclosed by the polar curve $r = f(\theta)$ ($\alpha \leq \theta \leq \beta$) and the lines $\theta = \alpha$ and $\theta = \beta$ is

$$A = \int_{\alpha}^{\beta} \frac{1}{2} [f(\theta)]^2 d\theta = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta \dots \dots \dots (3)$$



Area in Polar Coordinates: Limits of Integration

Step 1. **Sketch** the region R whose area is to be determined.

Step 2. Draw an arbitrary “radial line” from the pole to the boundary curve $r = f(\theta)$.

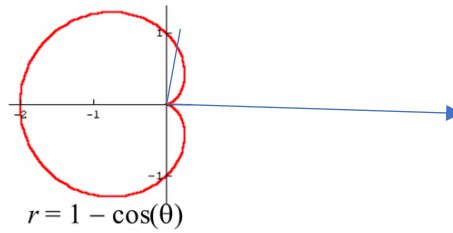
Step 3. Ask, “Over what interval of values must θ vary in order for the radial line to sweep out the region R ?”

Step 4. Your answer in Step 3 will determine the lower and upper limits of integration.

Exercise: 6

(a) Find the area of the region in the **first quadrant** that is within the cardioid $r = 1 - \cos\theta$.

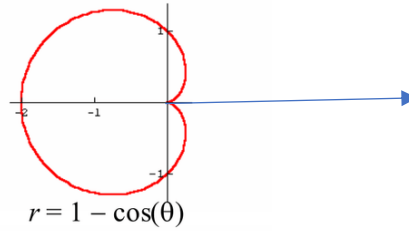




Given $r = 1 - \cos\theta$.

$$\begin{aligned}
 A &= \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta = \int_0^{\frac{\pi}{2}} \frac{1}{2} [1 - \cos\theta]^2 d\theta \\
 &= \int_0^{\frac{\pi}{2}} \frac{1}{2} [1 - 2\cos\theta + \cos^2\theta] d\theta \\
 &= \int_0^{\frac{\pi}{2}} \frac{1}{2} \left[1 - 2\cos\theta + \frac{1}{2}(1 + \cos(2\theta)) \right] d\theta \\
 &= \int_0^{\frac{\pi}{2}} \frac{1}{2} \left[\frac{2}{2} - \frac{4}{2}\cos\theta + \frac{1}{2}(1 + \cos(2\theta)) \right] d\theta \\
 &= \frac{1}{4} \int_0^{\frac{\pi}{2}} [2 - 4\cos\theta + 1 + \cos(2\theta)] d\theta \\
 &= \frac{1}{4} \int_0^{\frac{\pi}{2}} [3 - 4\cos\theta + \cos(2\theta)] d\theta \\
 &= \frac{1}{4} \left[3\theta - 4\sin\theta + \frac{1}{2}\sin(2\theta) \right]_0^{\frac{\pi}{2}} \\
 &= \frac{1}{4} \left(3 \cdot \frac{\pi}{2} - 4 \right) \\
 &= \frac{1}{8} (3\pi - 8) \text{ unit}^2
 \end{aligned}$$

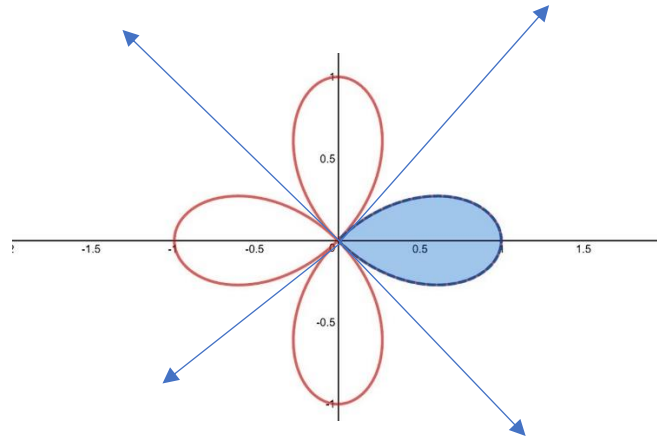
(b) Find the area of the region that is within the cardioid $r = 1 - \cos\theta$.



$$A = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta = \int_0^{2\pi} \frac{1}{2} [1 - \cos \theta]^2 d\theta = \frac{1}{4} \left[3\theta - 4 \sin \theta + \frac{1}{2} \sin(2\theta) \right]_0^{2\pi} = \frac{3\pi}{2} \text{ unit}^2$$

(c) Find the area of the region enclosed by the rose curve $r = \cos(2\theta)$.

Solution:



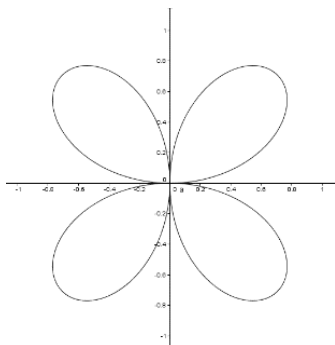
To find rays, set $r = 0$.

If $r = \cos(2\theta) = 0$; $[\cos(2n + 1) \frac{\pi}{2} = 0]$

$\Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$. Then $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$. If we want to take the area of the first petal, then the interval must be $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$, **since interval starts with smallest number and ends with the largest number.**

$$\begin{aligned} A &= \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta = 4 \times \text{Area of a petal} = 4 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{2} [\cos(2\theta)]^2 d\theta = 2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos^2(2\theta) d\theta \\ &= 2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{2} [1 + \cos(4\theta)] d\theta = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} [1 + \cos(4\theta)] d\theta = \left[\theta + \frac{1}{4} \sin(4\theta) \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = \frac{\pi}{2} \text{ unit}^2 \end{aligned}$$

(d) Find the area of the region enclosed by the rose curve $r = \sin(2\theta)$.

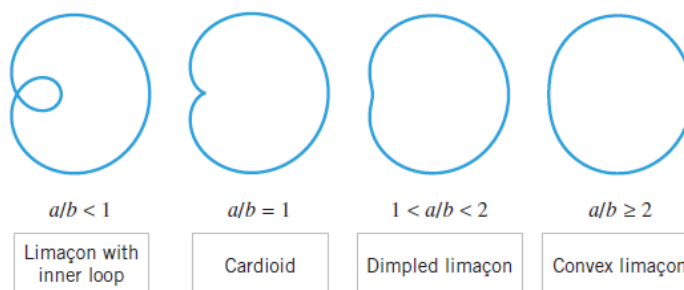


If $r = \sin 2\theta = 0$, then $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$. If we want to take the area of the petal in the first quadrant, then the interval must be $\left[0, \frac{\pi}{2}\right]$.

$$\begin{aligned}
 A &= 4 \int_0^{\frac{\pi}{2}} \frac{1}{2} [\sin(2\theta)]^2 d\theta = 2 \int_0^{\frac{\pi}{2}} \sin^2(2\theta) d\theta \\
 &= 2 \int_0^{\frac{\pi}{2}} \frac{1}{2} [1 - \cos(4\theta)] d\theta = \int_0^{\frac{\pi}{2}} [1 - \cos(4\theta)] d\theta = \left[\theta - \frac{1}{4} \sin(4\theta) \right]_0^{\frac{\pi}{2}} = \frac{\pi}{2} \text{ unit}^2
 \end{aligned}$$

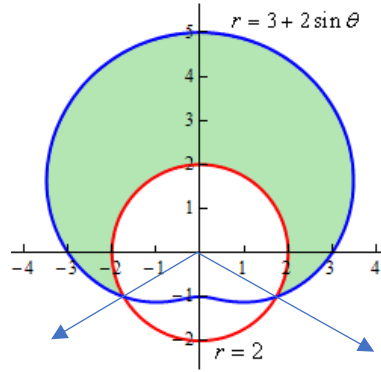
Limaçon Family:

$$r = a \pm b \sin \theta, \quad r = a \pm b \cos \theta$$



Exercise: 7

Find the area of the region that is inside the Dimpled limaçon $r = 3 + 2 \sin \theta$ outside the circle $r = 2$.



Given $r = 3 + 2 \sin \theta$ and $r = 2$.

To find the intersection:

$$\text{Set } 2 = 3 + 2 \sin \theta$$

$$\Rightarrow \sin \theta = -\frac{1}{2} ; \left[\sin \theta = -\frac{1}{2} \rightarrow \theta = \frac{\pi}{6} \right]$$

$$\therefore \theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6}, \text{ and } \theta = -\frac{\pi}{6}$$

[To move anti-clockwise **across the region**, $\theta = -\frac{\pi}{6}$]

$$\text{So, interval } \left[-\frac{\pi}{6}, \frac{7\pi}{6} \right]$$

Area of the region is on the interval $\left[-\frac{\pi}{6}, \frac{7\pi}{6} \right]$ is inside the Dimpled limaçon $r = 3 + 2 \sin \theta$ and outside the circle $r = 2$.

$$A = \text{Area of the Dimpled limaçon over the } \left[-\frac{\pi}{6}, \frac{7\pi}{6} \right] - \text{Area of the circle over the } \left[-\frac{\pi}{6}, \frac{7\pi}{6} \right]$$

$$= \int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} \frac{1}{2} [3 + 2 \sin \theta]^2 d\theta - \int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} \frac{1}{2} [2]^2 d\theta$$

$$= \frac{1}{2} \int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} [9 + 12 \sin \theta + 4 \sin^2 \theta] d\theta - \int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} 2 d\theta$$

$$= \frac{1}{2} \int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} \left[9 + 12 \sin \theta + 4 \cdot \frac{1}{2} (1 - \cos(2\theta)) \right] d\theta - \int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} 2 d\theta$$

$$\begin{aligned}
&= \frac{1}{2} \int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} [9 + 12 \sin \theta + 2 - 2 \cos(2\theta)] d\theta - \int_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} 2 d\theta \\
&= \frac{1}{2} \left[9\theta + 12(-\cos \theta) + 2\theta - 2 \frac{\sin(2\theta)}{2} \right]_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} - [2\theta]_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} \\
&= \frac{1}{2} \left[11\theta - 12 \cos \theta - \sin(2\theta) \right]_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} - [2\theta]_{-\frac{\pi}{6}}^{\frac{7\pi}{6}} \\
&= \frac{1}{2} \left[11 \left(\frac{7\pi}{6} \right) - 12 \cos \frac{7\pi}{6} - \sin \left(\frac{7\pi}{3} \right) - 11 \left(-\frac{\pi}{6} \right) + 12 \cos \left(-\frac{\pi}{6} \right) + \sin \left(-\frac{\pi}{3} \right) \right] \\
&\quad - \left[2 \left(\frac{7\pi}{6} \right) - 2 \left(-\frac{\pi}{6} \right) \right] \\
&= \frac{1}{2} \left[\frac{77\pi}{6} - 12 \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} + \frac{11\pi}{6} + 12 \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \right] - \frac{14\pi}{6} - \frac{2\pi}{6} \\
&= \frac{77\pi}{12} + \frac{11\pi}{12} - \frac{\sqrt{3}}{2} - \frac{7\pi}{3} - \frac{\pi}{3} \\
&= \frac{88 - 32}{12} \pi - \frac{\sqrt{3}}{2} \\
&= \frac{56}{12} \pi - \frac{\sqrt{3}}{2} \\
&= \frac{1}{12} [56\pi - 6\sqrt{3}] \text{ unit}^2
\end{aligned}$$

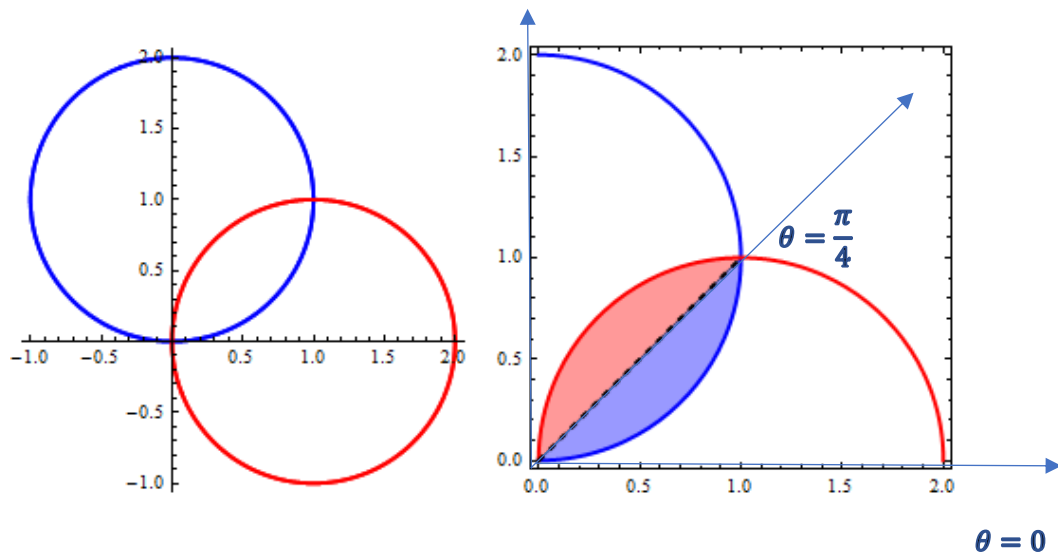
Exercise: 8

Find the Area of the polar region bounded by the circles $r = 2 \cos \theta$ and $r = 2 \sin \theta$.

Solution:

$$r = 2 \cos \theta \Rightarrow (x - 1)^2 + y^2 = 1^2 : \text{Center} = (1, 0), \text{Radius} = 1$$

$$\text{and } r = 2 \sin \theta \Rightarrow x^2 + (y - 1)^2 = 1^2 : \text{Center} = (0, 1) \text{ Radius} = 1$$



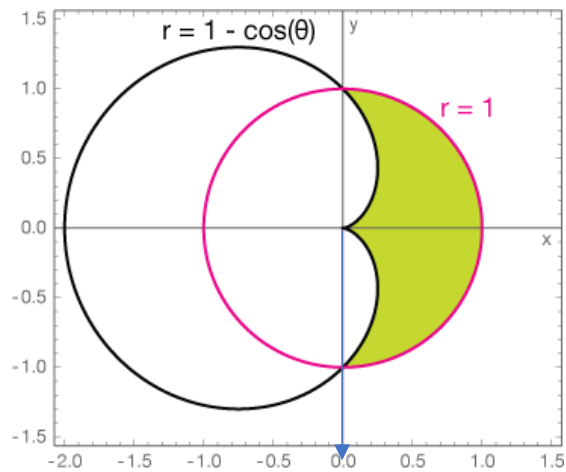
AREA = 2 × area of the bluish region

$$\begin{aligned}
 &= 2 \int_0^{\frac{\pi}{4}} \frac{1}{2} [2 \sin \theta]^2 d\theta \\
 &= 4 \int_0^{\frac{\pi}{4}} \frac{1}{2} [1 - \cos(2\theta)] d\theta \\
 &= 2 \left[\theta - \frac{\sin(2\theta)}{2} \right]_0^{\frac{\pi}{4}} \\
 &= \left(\frac{\pi}{2} - 1 \right) \text{ unit}^2
 \end{aligned}$$

Exercise: 9

Find the area of the polar region that is inside the circle $r = 1$ and outside the cardioid $r = 1 - \cos \theta$.





$$\text{Interval} = \left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

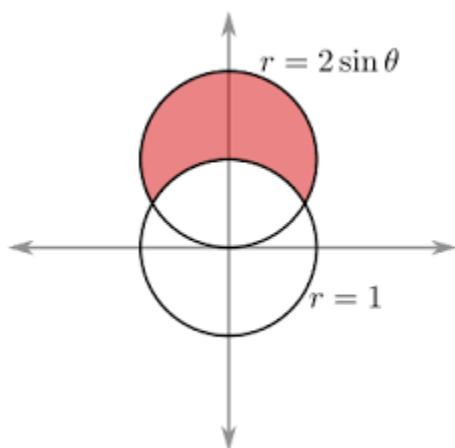
Area of the region = On the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$, Area of the Circle - area of the cardioid

$$= \frac{1}{2}\pi - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2}[1 - \cos \theta]^2 d\theta$$

Complete !!

Exercise: 10

Find the area of the polar region that is outside the circle $r = 1$ and outside the cardioid $r = 2 \sin \theta$.



Set $2 \sin \theta = 1$. So, $\sin \theta = \frac{1}{2}$.

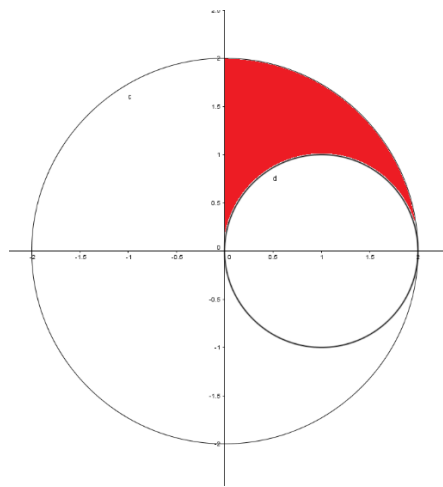
$$\text{Interval} = \left[\frac{\pi}{6}, \frac{5\pi}{6} \right]$$

Area= On the interval $\left[\frac{\pi}{6}, \frac{5\pi}{6} \right]$, area of the Circle $r = 2 \sin \theta$ — area of the circle $r = 1$

Exercise: 11

Find the area of the polar region in the first quadrant that is inside the circle $r = 2$ and outside the circle $r = 2 \cos \theta$.

$$r = 2 \cos \theta \Rightarrow (x - 1)^2 + y^2 = 1^2 : \text{Center} = (1, 0), \text{Radius} = 1$$



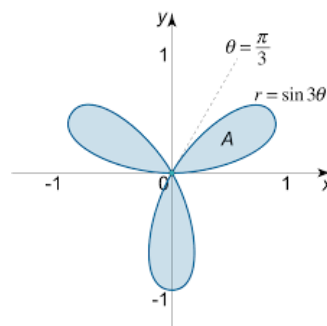
$$\text{Interval} = \left[0, \frac{\pi}{2} \right]$$

Area = On the interval $\left[0, \frac{\pi}{2} \right]$, area of the Circle $r = 2$ — area of the Circle $r = 2 \cos \theta$

Exercise: 12

Find the area of the rose $r = \sin 3\theta$.

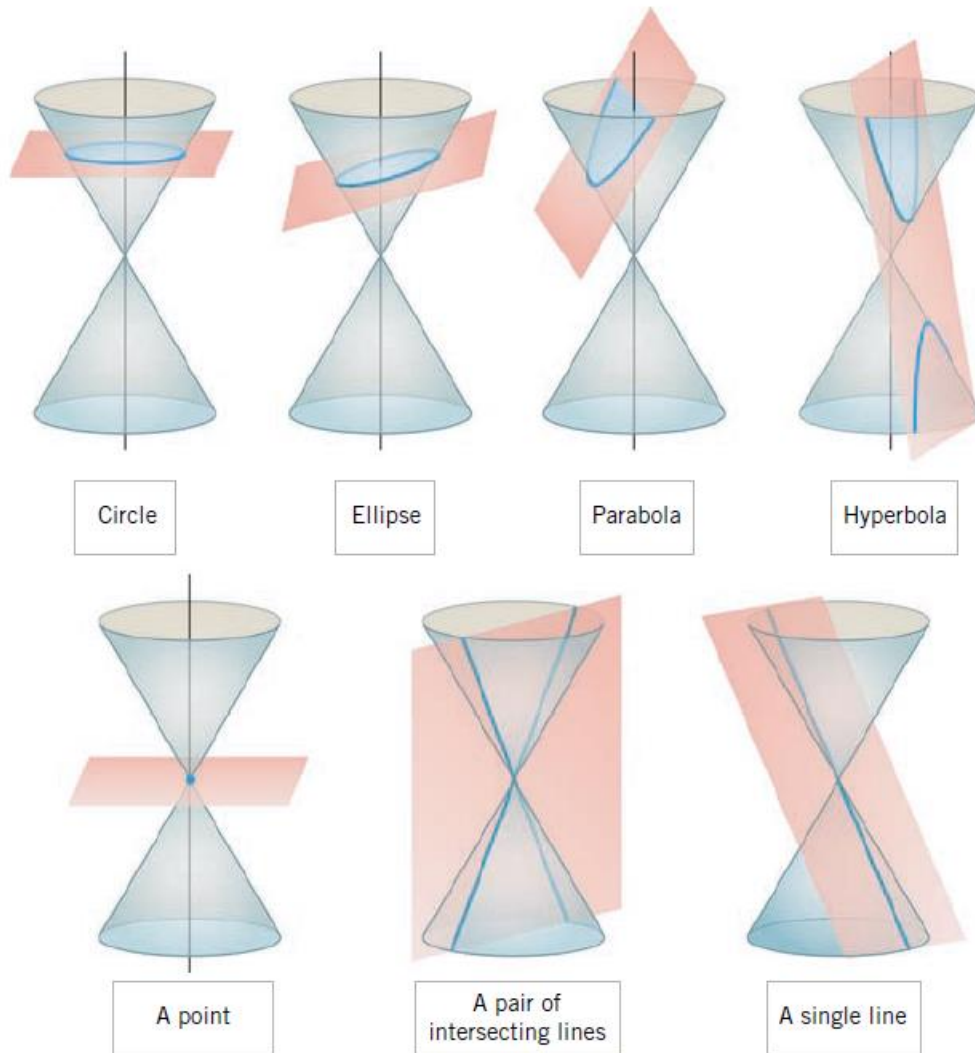
All yours!!!



Section: 10.6 Conic Sections in Polar Coordinates

CONIC SECTIONS

Circles, ellipses, parabolas, and hyperbolas are called *conic sections* or *conics* because they can be obtained as intersections of a plane with a double-napped circular cone (Figure 10.4.1). If the plane passes through the vertex of the double-napped cone, then the intersection is a point, a pair of intersecting lines, or a single line. These are called *degenerate conic sections*.



se

Recall: Write the equation in polar form: $x^2 = 4ay$, where a is a real number.

$$\text{Answer: } r = 4a \tan \theta \sec \theta = \frac{4a \sin \theta}{\cos^2 \theta} = \frac{4a \sin \theta}{(1 - \sin \theta)(1 + \sin \theta)}$$

Homework: Write the equation in polar form: $y^2 = 4bx$, where a is a real number.

Sections 10.4 and 10.6

Parabolas

DEFINITIONS Parabola, Focus, Directrix

A set that consists of all the points in a plane equidistant from a given fixed point and a given fixed line in the plane is a **parabola**. The fixed point is the **focus** of the parabola. The fixed line is the **directrix**.

TABLE 10.1 Standard-form equations for parabolas with vertices at the origin ($p > 0$)

Equation	Focus	Directrix	Axis	Opens
$x^2 = 4py$	$(0, p)$	$y = -p$	y -axis	Up
$x^2 = -4py$	$(0, -p)$	$y = p$	y -axis	Down
$y^2 = 4px$	$(p, 0)$	$x = -p$	x -axis	To the right
$y^2 = -4px$	$(-p, 0)$	$x = p$	x -axis	To the left

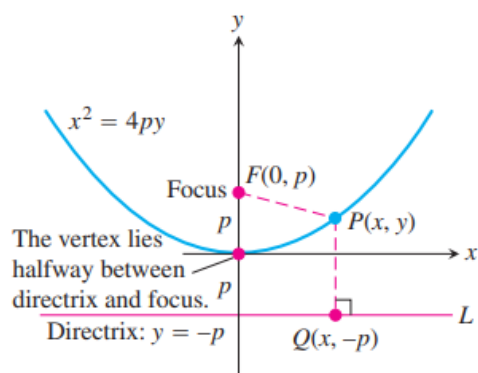


FIGURE 10.2 The standard form of the parabola $x^2 = 4py$, $p > 0$.

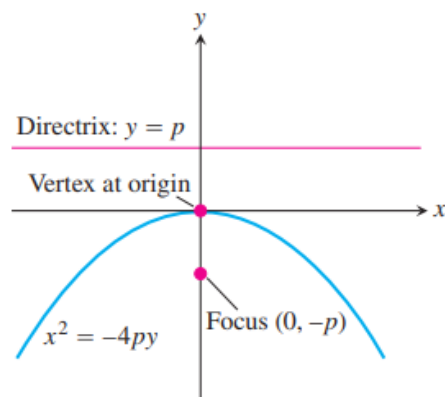


FIGURE 10.3 The parabola $x^2 = -4py$, $p > 0$.

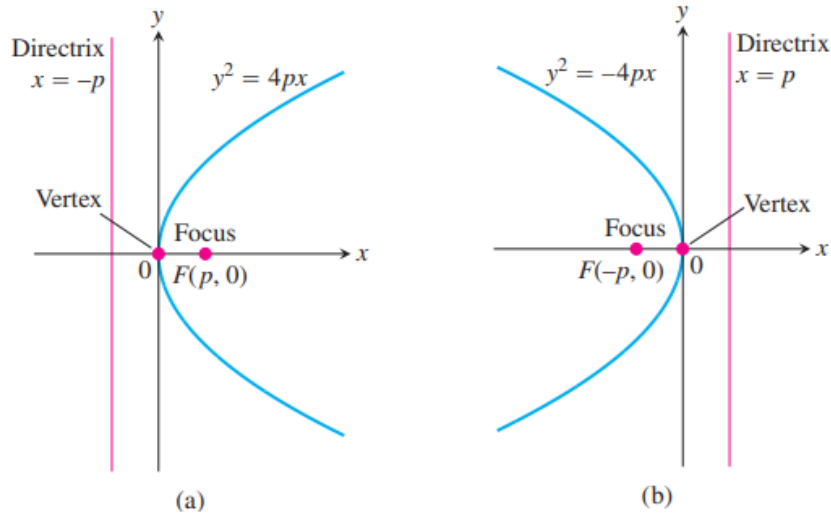


FIGURE 10.4 (a) The parabola $y^2 = 4px$. (b) The parabola $y^2 = -4px$.

Observations: $x^2 = -4py$ is Symmetric with respect to y-axis. Opening downward.

Axis of the parabola: $x^2 = 0 \Rightarrow x = 0$; y -axis.

Focus = (x coordinate will be on the axis of the parabola, y - coordinate $y = -p$)

Equation of the directrix: $y = p$

Example 1: Find the axis, focus, directrix and the vertex of the parabola given below:

$$(x + 2)^2 = -3(y - 4).$$

Solution:

Set, $X = x + 2$, $Y = y - 4$: $X^2 = -3Y$

$$\therefore X^2 = -4 \cdot \frac{3}{4} \cdot Y ; \text{ Compare: } x^2 = -4py; \quad p = \frac{3}{4}.$$

The parabola is opening downward.

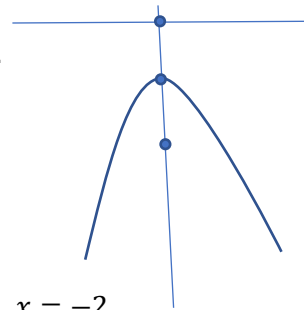
Axis: Set $X^2 = 0 \rightarrow X = 0 \rightarrow x + 2 = 0 \Rightarrow x = -2$.

The parabola is symmetric with respect to its axis $x = -2$.

Focus: x-coordinate must be on the axis of the parabola, that is, $x = -2$

and y-coordinate comes from the equation $Y = -p \rightarrow y - 4 = -\frac{3}{4} \Rightarrow y = \frac{13}{4}$

$$\therefore \text{Focus} = \left(-2, \frac{13}{4}\right).$$



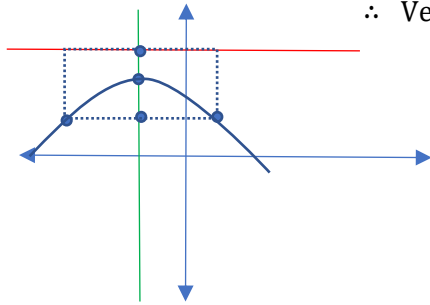
Directrix: $Y = p \rightarrow y - 4 = \frac{3}{4} \therefore \text{Line: } y = \frac{19}{4}$

What is the vertex?

Vertex = Midpoint of $\left(-2, \frac{19}{4}\right)$ and [intersection of the x -axis $x = -2$ and the directrix $y = \frac{19}{4}$]

= Midpoint of $\left(-2, \frac{13}{4}\right)$ and $\left(-2, \frac{19}{4}\right) = \left(\frac{-2-2}{2}, \frac{\frac{13}{4} + \frac{19}{4}}{2}\right)$

$\therefore \text{Vertex} = (-2, 4)$



Example2:(A) Find the focus and directrix of the parabola $3y^2 = 4x$.

Here given parabola $3y^2 = 4x \rightarrow y^2 = \frac{4}{3}x$

$\rightarrow y^2 = 4 \cdot \frac{1}{3} \cdot x ; y^2 = 4px$

$\therefore p = \frac{1}{3} > 0$.

For

$y^2 = 4px \quad Y^2 = 4pX$

Opening to the right

Axis of the parabola $y^2 = 4px : y^2 = 0 \rightarrow y = 0 \rightarrow x$ -axis.

Focus $(p, 0) = \left(\frac{1}{3}, 0\right) ; F = (x \text{ coordinate comes from } x=p, y \text{ coordinate on the axis of the parabole})$

Equation of the directrix: vertical line. Equation $X = -p, x = -\frac{1}{3}$.

What is the vertex: Midpoint of $\left(\frac{1}{3}, 0\right)$ and [intersection of the x -axis and the directrix]

= Midpoint of $\left(\frac{1}{3}, 0\right)$ and $\left(-\frac{1}{3}, 0\right) = \left(\frac{\frac{1}{3} - \frac{1}{3}}{2}, \frac{0}{2}\right) = (0, 0)$

Example 2: (B) Find the focus and directrix of the parabola $3(y + 2)^2 = 4(x - 1)$.

Set $Y = y + 2, \quad X = x - 1, \text{ then } Y^2 = 4 \cdot \frac{1}{3} \cdot X ; \quad y^2 = 4px$

- ➔ The parabola is opening to the right
- ➔ Axis of the parabola: $Y = 0 \rightarrow y + 2 = 0 \quad \therefore y = -2$.
Here, $p = \frac{1}{3}$

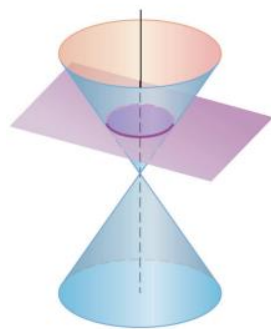
Focus = a point on the axis of the parabola where $X = p \rightarrow x - 1 = \frac{1}{3}$. $F = \left(\frac{4}{3}, -2\right)$

Equation of the directrix: vertical line. Equation $X = -p$, $x - 1 = -\frac{1}{3} \quad \therefore \text{Line } x = \frac{2}{3}$.

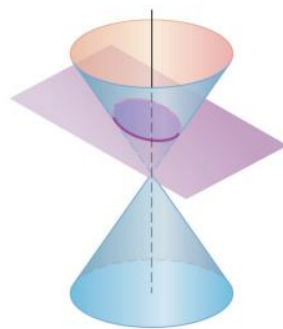
What is the vertex: Midpoint of $\left(\frac{1}{3}, 0\right)$ and [intersection of the x -axis and the directrix]

= Midpoint of $\left(\frac{4}{3}, -2\right)$ and $\left(\frac{2}{3}, -2\right) = \left(\frac{\frac{4}{3} + \frac{2}{3}}{2}, \frac{-2 - 2}{2}\right) = (1, -2)$.

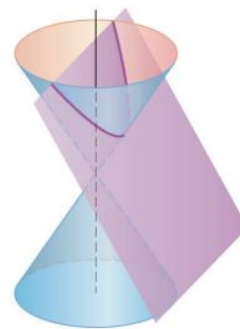
Conic Sections



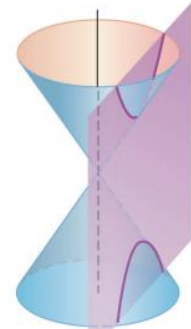
Circle: plane perpendicular to cone axis



Ellipse: plane oblique to cone axis



Parabola: plane parallel to side of cone



Hyperbola: plane cuts both halves of cone

(a)

Ellipses

DEFINITIONS Ellipse, Foci

An **ellipse** is the set of points in a plane whose distances from two fixed points in the plane have a constant sum. The two fixed points are the **foci** of the ellipse.

DEFINITIONS Focal Axis, Center, Vertices

The line through the foci of an ellipse is the ellipse's **focal axis**. The point on the axis halfway between the foci is the **center**. The points where the focal axis and ellipse cross are the ellipse's **vertices** (Figure 10.6).

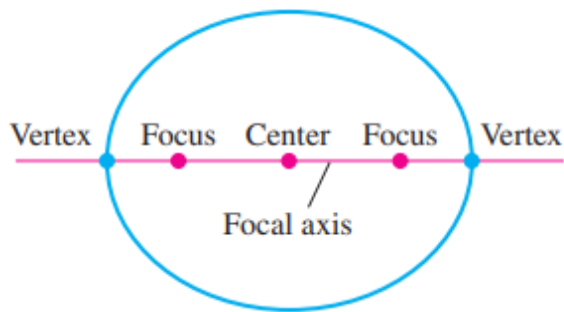


FIGURE 10.6 Points on the focal axis of an ellipse.

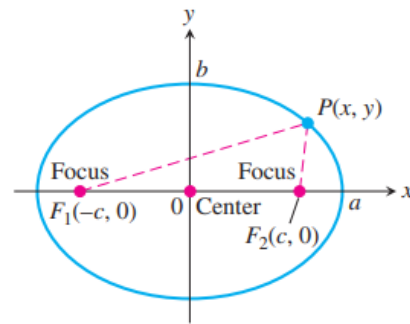


FIGURE 10.7 The ellipse defined by the equation $PF_1 + PF_2 = 2a$ is the graph of the equation $(x^2/a^2) + (y^2/b^2) = 1$, where $b^2 = a^2 - c^2$.

➔ Major, minor axes, vertices, Foci, center.

Equations in standard form: $a > b > 0$:

Case: 1 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Major axis: x -axis, Equation of the major axis $y = 0$.

Minor axis: y -axis, Equation of the major axis $x = 0$.

Vertices: These points are on the major axis $y=0$. Set $y=0$ in

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \rightarrow x^2 = a^2 \quad \therefore x = \pm a$$

Vertices: $(-a, 0), (a, 0)$

Center: The midpoint of the vertices (of the foci). (intersection of the major and minor axis)

Center = $(0,0)$.

Foci are on the Major axis given by $(\pm c, 0)$. Note: $x = \pm c$ and $y = 0$.

where $c = \sqrt{a^2 - b^2}$.

Case -II $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

Major axis: y –axis, Equation of the major axis $x = 0$.

Minor axis: x –axis, Equation of the major axis $y = 0$.

Verticas: These points are on the major axis $x = 0$. Set $x = 0$ in

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \rightarrow y^2 = a^2 \quad \therefore y = \pm a$$

Vertices: $(0, -a), (0, a)$

Center: The midpoint of the vertices (of the foci). (intersection of the major and minor axis)

Center = $(0,0)$.

Foci are on the Major axis given by $(0, \pm c)$. Note: $x = 0$ and $y = \pm c$.

where $c = \sqrt{a^2 - b^2}$.

Example:

$$9x^2 + 4y^2 = 36 \rightarrow \frac{9x^2}{36} + \frac{4y^2}{36} = 1 \rightarrow$$

$$\therefore \frac{x^2}{2^2} + \frac{y^2}{3^2} = 1 ; \left[\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \right]$$

Length of Major axis $>$ Length of Minor axis.

Observation: (Ellipse in Standard form)

There are two line of symetries which are x-axis and y-axis.

$$x - \text{axis: } y = 0 \rightarrow \frac{x^2}{2^2} = 1 \rightarrow x^2 = 2^2 \quad \therefore x = \pm 2 \rightarrow \text{Points: } (2,0) \text{ \& } (-2,0)$$

$$y - \text{axis: } x = 0 \rightarrow \frac{y^2}{3^2} = 1 \rightarrow y^2 = 3^2 \quad \therefore y = \pm 3 \rightarrow \text{Points: } (0,3) \text{ \& } (0,-3)$$

Ends of the major axis are called the vertices: $(0,3)$ & $(0,-3)$.

Major axis: y – axis

Minor axis: x – axis.

Foci are always on the major axis. Foci are: $(0, \pm c)$; where $c = \sqrt{a^2 - b^2} = \sqrt{9 - 4} = \sqrt{5}$.

Hence, Foci are: $(0, \pm\sqrt{5})$

Center $(0,0)$.

Example: Graph the Ellipse given by $\frac{(x-3)^2}{5^2} + \frac{(y+2)^2}{3^2} = 1$.

Set $X = x - 3$, $Y = y + 2$. Then, $\frac{X^2}{5^2} + \frac{Y^2}{3^2} = 1$.

[Axes of $\frac{X^2}{5^2} + \frac{Y^2}{3^2} = 1$: Set $Y = 0$ and solve X and set $X = 0$ and solve for Y .]

Equation of the major axis: Set $Y = 0 \rightarrow y + 2 = 0 \therefore y = -2$.

Vertices are on the major axis given by: If $Y = 0$, then $\frac{X^2}{5^2} = 1$

$$\rightarrow X^2 = 5^2 \rightarrow (x - 3)^2 = 5^2 \rightarrow x - 3 = \pm 5 \therefore x = 8, -2$$

Vertices: $(8, -2)$ and $(-2, -2)$

Center = Midpoint of the vertices = $\left(\frac{8-2}{2}, \frac{-2-2}{2}\right) = (3, -2)$.

Equation of the minor axis: Set $X = 0 \rightarrow x - 3 = 0 \therefore x = 3$.

The points on the minor axis: If $X = 0$, then

$$\frac{Y^2}{3^2} = 1 \rightarrow Y^2 = 3^2 \rightarrow (y + 2)^2 = 3^2 \rightarrow y + 2 = \pm 3 \therefore y = 1, -5$$

Points on the minor axis are: $(3, 1)$ and $(3, -5)$

Now, distance between a focus and the center:

$$c = \sqrt{a^2 - b^2} = \sqrt{5^2 - 3^2} = 4$$

The Foci are on the major axis, so we get $X = \pm c$ and $Y = 0$.

$$\rightarrow x - 3 = \pm 4 \text{ and } y + 2 = 0 \rightarrow x = 7, -1 \text{ and } y = -2.$$

Foci: $(7, -2)$ and $(-1, -2)$

Sketch????

....

DONE

One axis: $Y = (y + 2) = 0 \therefore y = -2$. Now, if $Y = 0$, then

$$(x - 3)^2 = 5^2 \rightarrow x - 3 = \pm 5 \therefore x = 8, -2$$

Major axis: $y = -2$.

Other axis: $X = (x - 3) = 0 \therefore x = 3$. Now, if $X = 0$, then

$$(y + 2)^2 = 3^2 \rightarrow y + 2 = \pm 3 \therefore y = 1, -5$$

Minor Axis: $x = 3$

$$\text{Center} \left(\frac{(8-2)}{2}, \frac{-2-2}{2} \right) = (3, -2)$$

Foci are always on the major axis where $c = \sqrt{a^2 - b^2} = \sqrt{5^2 - 3^2} = 4$.

For Foci, $X = \pm c$ and $Y = 0 \rightarrow x - 3 = \pm 4 \therefore x = 7, -1$.

Foci are on the major axis: $y = -2$

Hence, Foci are: $(X, Y) = (\pm c, 0)$

Foci are: $(7, -2)$ and $(-1, -2)$

Standard-Form Equations for Ellipses Centered at the Origin

Foci on the x-axis: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b)$

Center-to-focus distance: $c = \sqrt{a^2 - b^2}$

Foci: $(\pm c, 0)$

Vertices: $(\pm a, 0)$

Foci on the y-axis: $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \quad (a > b)$

Center-to-focus distance: $c = \sqrt{a^2 - b^2}$

Foci: $(0, \pm c)$

Vertices: $(0, \pm a)$

In each case, a is the semimajor axis and b is the semiminor axis.

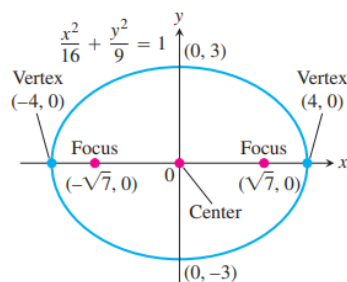


FIGURE 10.8 An ellipse with its major axis horizontal (Example 2).

EXAMPLE 2 Major Axis Horizontal

The ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

(Figure 10.8) has

Semimajor axis: $a = \sqrt{16} = 4$, Semiminor axis: $b = \sqrt{9} = 3$

Center-to-focus distance: $c = \sqrt{16 - 9} = \sqrt{7}$

Foci: $(\pm c, 0) = (\pm\sqrt{7}, 0)$

Vertices: $(\pm a, 0) = (\pm 4, 0)$

Center: $(0, 0)$.

EXAMPLE 3 Major Axis Vertical

The ellipse

$$\frac{x^2}{9} + \frac{y^2}{16} = 1, \quad (6)$$

obtained by interchanging x and y in Equation (5), has its major axis vertical instead of horizontal (Figure 10.9). With a^2 still equal to 16 and b^2 equal to 9, we have

Semimajor axis: $a = \sqrt{16} = 4$, Semiminor axis: $b = \sqrt{9} = 3$

Center-to-focus distance: $c = \sqrt{16 - 9} = \sqrt{7}$

Foci: $(0, \pm c) = (0, \pm\sqrt{7})$

Vertices: $(0, \pm a) = (0, \pm 4)$

Center: $(0, 0)$.

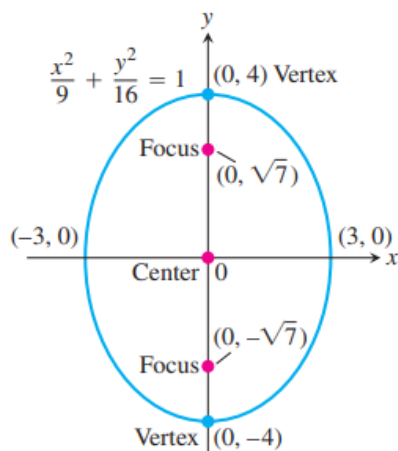
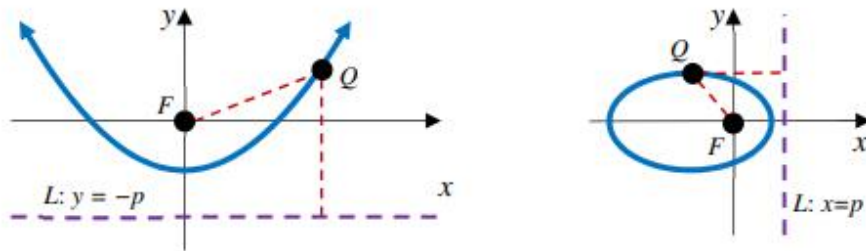


FIGURE 10.9 An ellipse with its major axis vertical (Example 3).

Conic Sections in Polar Coordinates

Conic Sections General Definition

A **conic section** can be defined by placing a fixed point at the origin, $F(0,0)$, called the **focus**, and drawing a line L called the **directrix** at $x = \pm p$ or $y = \pm p$. The conic section is the set of all points $Q(x, y)$ for which the ratio of the distance from Q to F to the distance from Q to the directrix is some positive constant e , called the **eccentricity**. In other words, $\frac{d(Q, F)}{d(Q, L)} = e$.



Warning: the eccentricity, e , is **not** the Euler constant $e \approx 2.71828$ we studied with exponentials

Polar Equation for a Conic Section

A **conic section** with a focus at the origin, **eccentricity** e , and **directrix** at $x = \pm p$ or $y = \pm p$ will have polar equation:

$$r = \frac{ep}{1 \pm e \sin(\theta)} \quad \text{when the directrix is } y = \pm p$$

$$r = \frac{ep}{1 \pm e \cos(\theta)} \quad \text{when the directrix is } x = \pm p$$

Notes: In standard form, conic sections are either symmetric with respect to x –axis (Directrix is vertical) or with respect to y –axis (Directrix is Horizontal).

Focus of the Conic section is at the origin , $F = (0,0)$. Say, $p > 0$.

Directrix	Conic Section
$x = p$	$r = \frac{ep}{1 + e \cos \theta}$
$x = -p$	$r = \frac{ep}{1 - e \cos \theta}$
$y = p$	$r = \frac{ep}{1 + e \sin \theta}$
$y = -p$	$r = \frac{ep}{1 - e \sin \theta}$

Find the equation of the Conic section with directrix $x = 3$ & eccentricity $e = 2$.

Note: Since $e = 2 > 1$, so it is a hyperbola. Here $p=3$. $E=2$

Directrix is vertical $\rightarrow$ The conic section is perpendicular to x-axis.

Equation: $r = \frac{ep}{1+e \cos \theta} \quad \therefore \quad r = \frac{6}{1+2 \cos \theta}$

Relation Between the Polar Equation of a Conic and its Shape

For a **conic section** with a focus at the origin, **eccentricity** e , and **directrix** at $x = \pm p$ or $y = \pm p$,

when $0 < e < 1$, the graph is an ellipse

when $e = 1$, the graph is a parabola

when $e > 1$, the graph is a hyperbola

Example: What is the shape of the conic section? Find the Directrix.

$$r = \frac{3}{5 - 2 \sin \theta}.$$

Solution: The form of the conic section must be $r = \frac{ep}{1 \pm e \sin \theta}$.

$$\text{Now, } r = \frac{3}{5 - 2 \sin \theta} = r = \frac{3}{5(1 - \frac{2}{5} \sin \theta)}$$

$$\therefore r = \frac{\frac{3}{5}}{1 - \frac{2}{5} \sin \theta} = \frac{ep}{1 - e \sin \theta}$$

Here, eccentricity: $e = \frac{2}{5}$. Since $0 < e < 1$, so the conic section is an Ellipse.

Also,

$$ep = \frac{3}{5} \rightarrow \frac{2}{5} p = \frac{3}{5} \quad \therefore p = \frac{3}{2}.$$

$$\text{Directrix: } y = -\frac{3}{2}.$$

.....

One of the vertex: Midpoint between the (0,0) and the line $y = -\frac{3}{2}$

The major axis is the y-axis. $x = 0$.

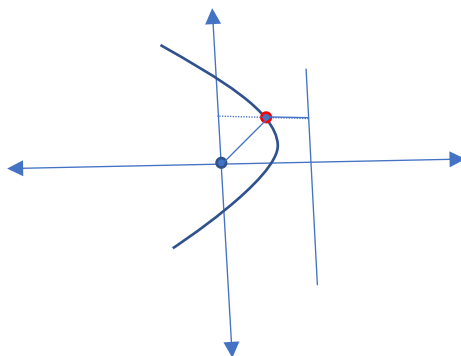
Vertex $(-\frac{3}{4}, 0)$

Important: Graphing Conic sections (parabola and Ellipse) in polar coordinate.

Important Topics of This Section
Polar equations for Conic Sections
Eccentricity and Directrix
Determining the shape of a polar conic section

**** Derive the formula for the **parabola** with focus $F = (0,0)$ and directrix $x = p$.

Parabola in Polar coordinates:



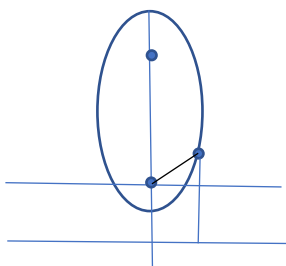
Fixed point Focus $F = (0,0)$ and Fixed line Directrix line $L: x = p$.

Consider an arbitrary point $Q(x, y) = (r, \theta)$. By definition

$$\frac{d(Q, F)}{d(Q, L)} = 1 \rightarrow \frac{r}{p - x} = 1 \rightarrow \frac{r}{p - r \cos \theta} = 1$$

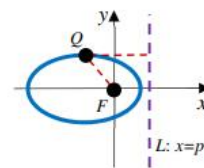
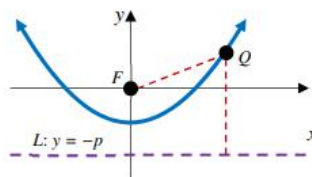
$$\rightarrow r = p - r \cos \theta \rightarrow r + r \cos \theta = p$$

$$\therefore r = \frac{p}{1 + \cos \theta}$$



Conic Sections General Definition

A **conic section** can be defined by placing a fixed point at the origin, $F(0,0)$, called the **focus**, and drawing a line L called the **directrix** at $x = \pm p$ or $y = \pm p$. The conic section is the set of all points $Q(x, y)$ for which the ratio of the distance from Q to F to the distance from Q to the directrix is some positive constant e , called the **eccentricity**. In other words, $\frac{d(Q, F)}{d(Q, L)} = e$.



Warning: the eccentricity, e , is **not** the Euler constant $e \approx 2.71828$ we studied with exponentials

Ellipse: $Q = (x, y)$, Focus $F = (0,0)$, Directrix Line $L: y = -p$

Parabola: $Q = (x, y)$, Focus $F = (0,0)$, Directrix Line $L: y = -p$

$$\frac{d(Q, F)}{d(Q, L)} = e ;$$

[$e = 1$ for parabola and for ellipse $0 < e < 1$]

In polar coordinate: $\frac{d(Q, F)}{d(Q, L)} = e \rightarrow \frac{r}{y+p} = e$

$$\rightarrow \frac{r}{r \sin \theta + p} = e \rightarrow r = er \sin \theta + ep$$

$$\rightarrow r - er \sin \theta = ep$$

$$\therefore r = \frac{ep}{1 - e \sin \theta}$$

Sections 10.4 and 10.6

Parabolas

DEFINITIONS Parabola, Focus, Directrix

A set that consists of all the points in a plane equidistant from a given fixed point and a given fixed line in the plane is a **parabola**. The fixed point is the **focus** of the parabola. The fixed line is the **directrix**.

TABLE 10.1 Standard-form equations for parabolas with vertices at the origin ($p > 0$)

Equation	Focus	Directrix	Axis	Opens
$x^2 = 4py$	$(0, p)$	$y = -p$	y -axis	Up
$x^2 = -4py$	$(0, -p)$	$y = p$	y -axis	Down
$y^2 = 4px$	$(p, 0)$	$x = -p$	x -axis	To the right
$y^2 = -4px$	$(-p, 0)$	$x = p$	x -axis	To the left

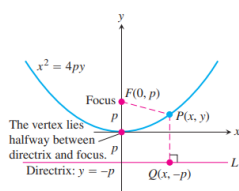


FIGURE 10.2 The standard form of the parabola $x^2 = 4py$, $p > 0$.

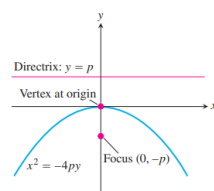


FIGURE 10.3 The parabola $x^2 = -4py$, $p > 0$.

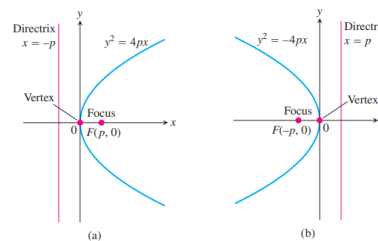


FIGURE 10.4 (a) The parabola $y^2 = 4px$. (b) The parabola $y^2 = -4px$.

Observations: $x^2 = -4py$ is Symmetric with respect to y -axis. Opening downward.

Axis of the parabola: $x^2 = 0 \therefore x = 0$; y -axis.

Focus = (x coordinate will be on the axis of the parabola, y - coordinate $y = -p$)

Equation of the directrix: $y = p$

Example 1: Find the axis, focus, directrix and the vertex of the parabola given below:

$$(x + 2)^2 = -3(y - 4).$$

Solution:

Set, $X = x + 2$, $Y = y - 4$: $X^2 = -3Y$

$$\therefore X^2 = -4 \cdot \frac{3}{4} \cdot Y ; \text{ Compare: } x^2 = -4py; \quad p = \frac{3}{4}.$$

The parabola is opening downward.

Axis: Set $X^2 = 0 \rightarrow X = 0 \rightarrow x + 2 = 0 \therefore x = -2$.

The parabola is symmetric with respect to its axis $x = -2$.

Focus: x-coordinate must be on the axis of the parabola, that is, $x = -2$

and y-coordinate comes from the equation $Y = -p \rightarrow y - 4 = -\frac{3}{4} \therefore y = \frac{13}{4}$

$$\therefore \text{Focus} = \left(-2, \frac{13}{4}\right).$$

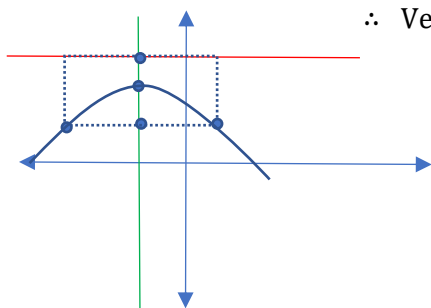
Directrix: $Y = p \rightarrow y - 4 = \frac{3}{4} \therefore \text{Line: } y = \frac{19}{4}$

What is the vertex?

Vertex=Midpoint of $\left(-2, \frac{19}{4}\right)$ and [intersection of the x -axis $x = -2$ and the directrix $y = \frac{19}{4}$]

$$=\text{Midpoint of } \left(-2, \frac{13}{4}\right) \text{ and } \left(-2, \frac{19}{4}\right) = \left(\frac{-2-2}{2}, \frac{\frac{13}{4}+\frac{19}{4}}{2}\right)$$

$$\therefore \text{Vertex} = (-2, 4)$$



Example2:(A) Find the focus and directrix of the parabola $3y^2 = 4x$.

Here given parabola $3y^2 = 4x \rightarrow y^2 = \frac{4}{3}x$

$$\rightarrow y^2 = 4 \cdot \frac{1}{3} \cdot x ; \quad y^2 = 4px$$

$$\therefore p = \frac{1}{3} > 0.$$

For

$$y^2 = 4px \quad Y^2 = 4pX$$

Opening to the right

Axis of the parabola $y^2 = 4px : y^2 = 0 \rightarrow y = 0 \rightarrow x\text{-axis}.$

Focus $(p, 0) = \left(\frac{1}{3}, 0\right)$; $F=(x \text{ coordinate comes from } x=p, y \text{ coordinate on the axis of the parabole})$

Equation of the directrix: vertical line. Equation $X = -p$, $x = -\frac{1}{3}.$

What is the vertex: Midpoint of $\left(\frac{1}{3}, 0\right)$ and [intersection of the x -axis and the directrix]

$$= \text{Midpoint of } \left(\frac{1}{3}, 0\right) \text{ and } \left(-\frac{1}{3}, 0\right) = \left(\frac{\frac{1}{3}-\frac{1}{3}}{2}, \frac{0}{2}\right) = (0,0)$$

Example 2: (B) Find the focus and directrix of the parabola $3(y+2)^2 = 4(x-1).$

$$\text{Set } Y = y + 2, \quad X = x - 1, \text{ then } Y^2 = 4 \cdot \frac{1}{3} \cdot X ; \quad y^2 = 4px$$

➔ The parabola is opening to the right

➔ Axis of the parabola: $Y = 0 \rightarrow y + 2 = 0 \quad \therefore y = -2.$

$$\text{Here, } p = \frac{1}{3}$$

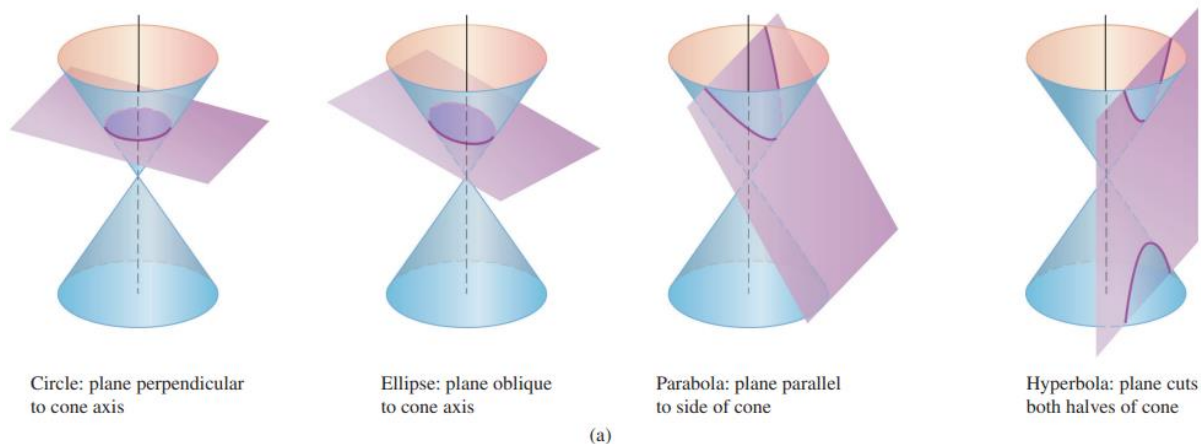
Focus = a point on the axis of the parabola where $X = p \rightarrow x - 1 = \frac{1}{3}. \quad F = \left(\frac{4}{3}, -2\right)$

Equation of the directrix: vertical line. Equation $X = -p$, $x - 1 = -\frac{1}{3} \quad \therefore \text{Line } x = \frac{2}{3}.$

What is the vertex: Midpoint of $\left(\frac{1}{3}, 0\right)$ and [intersection of the x -axis and the directrix]

$$= \text{Midpoint of } \left(\frac{4}{3}, -2\right) \text{ and } \left(\frac{2}{3}, -2\right) = \left(\frac{\frac{4}{3}+\frac{2}{3}}{2}, \frac{-2-2}{2}\right) = (1, -2).$$

Conic Sections



Ellipses

DEFINITIONS Ellipse, Foci

An **ellipse** is the set of points in a plane whose distances from two fixed points in the plane have a constant sum. The two fixed points are the **foci** of the ellipse.

DEFINITIONS Focal Axis, Center, Vertices

The line through the foci of an ellipse is the ellipse's **focal axis**. The point on the axis halfway between the foci is the **center**. The points where the focal axis and ellipse cross are the ellipse's **vertices** (Figure 10.6).

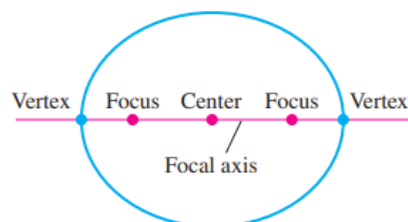


FIGURE 10.6 Points on the focal axis of an ellipse.

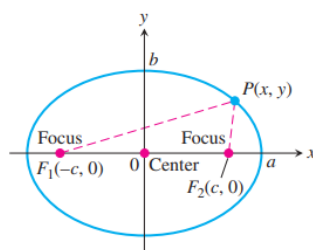


FIGURE 10.7 The ellipse defined by the equation $PF_1 + PF_2 = 2a$ is the graph of the equation $(x^2/a^2) + (y^2/b^2) = 1$, where $b^2 = a^2 - c^2$.

➔ Major, minor axes, vertices, Foci, center.

Equations in standard form: $a > b > 0$:

Case: 1 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Major axis: x –axis, Equation of the major axis $y = 0$.

Minor axis: y –axis, Equation of the major axis $x = 0$.

Vertices: These points are on the major axis $y=0$. Set $y=0$ in

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \rightarrow x^2 = a^2 \quad \therefore x = \pm a$$

Vertices: $(-a, 0), (a, 0)$

Center: The midpoint of the vertices (of the foci). (intersection of the major and minor axis)

Center = $(0,0)$.

Foci are on the Major axis given by $(\pm c, 0)$. Note: $x = \pm c$ and $y = 0$.

where $c = \sqrt{a^2 - b^2}$.

Case -II $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

Major axis: y –axis, Equation of the major axis $x = 0$.

Minor axis: x –axis, Equation of the major axis $y = 0$.

Vertices: These points are on the major axis $x = 0$. Set $x = 0$ in

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \rightarrow y^2 = a^2 \quad \therefore y = \pm a$$

Vertices: $(0, -a), (0, a)$

Center: The midpoint of the vertices (of the foci). (intersection of the major and minor axis)

Center = $(0,0)$.

Foci are on the Major axis given by $(0, \pm c)$. Note: $x = 0$ and $y = \pm c$.

where $c = \sqrt{a^2 - b^2}$.

Example:

$$9x^2 + 4y^2 = 36 \rightarrow \frac{9x^2}{36} + \frac{4y^2}{36} = 1 \rightarrow$$

$$\therefore \frac{x^2}{2^2} + \frac{y^2}{3^2} = 1 ; \left[\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \right]$$

Length of Major axis > Length of Minor axis.

Observation: (Ellipse in Standard form)

There are two line of symetries which are x-axis and y-axis.

$$x - \text{axis: } y = 0 \rightarrow \frac{x^2}{2^2} = 1 \rightarrow x^2 = 2^2 \therefore x = \pm 2 \rightarrow \text{Points: } (2,0) \text{ \& } (-2,0)$$

$$y - \text{axis: } x = 0 \rightarrow \frac{y^2}{3^2} = 1 \rightarrow y^2 = 3^2 \therefore y = \pm 3 \rightarrow \text{Points: } (0,3) \text{ \& } (0,-3)$$

Ends of the major axis are called the vertices: $(0,3)$ & $(0,-3)$.

Major axis: y – axis

Minor axis: x – axis.

Foci are always on the major axis. Foci are: $(0, \pm c)$; where $c = \sqrt{a^2 - b^2} = \sqrt{9 - 4} = \sqrt{5}$.

Hence, Foci are: $(0, \pm\sqrt{5})$

Center $(0,0)$.

Example: Graph the Ellipse given by $\frac{(x-3)^2}{5^2} + \frac{(y+2)^2}{3^2} = 1$.

$$\text{Set } X = x - 3, \quad Y = y + 2. \text{ Then, } \frac{X^2}{5^2} + \frac{Y^2}{3^2} = 1.$$

[Axes of $\frac{X^2}{5^2} + \frac{Y^2}{3^2} = 1$: Set $Y = 0$ and solve X and set $X = 0$ and solve for Y .]

Equation of the major axis: Set $Y = 0 \rightarrow y + 2 = 0 \therefore y = -2$.

Vertices are on the major axis given by: If $Y = 0$, then $\frac{X^2}{5^2} = 1$

$$\rightarrow X^2 = 5^2 \rightarrow (x - 3)^2 = 5^2 \rightarrow x - 3 = \pm 5 \therefore x = 8, -2$$

Vertices: $(8, -2)$ and $(-2, -2)$

Center = Midpoint of the vertices = $\left(\frac{8-2}{2}, \frac{-2-2}{2} \right) = (3, -2)$.

Equation of the minor axis: Set $X = 0 \rightarrow x - 3 = 0 \therefore x = 3$.

The points on the minor axis: If $X = 0$, then

$$\frac{Y^2}{3^2} = 1 \rightarrow Y^2 = 3^2 \rightarrow (y + 2)^2 = 3^2 \rightarrow y + 2 = \pm 3 \therefore y = 1, -5$$

Points on the minor axis are: $(3, 1)$ and $(3, -5)$

Now, distance between a focus and the center:

$$c = \sqrt{a^2 - b^2} = \sqrt{5^2 - 3^2} = 4$$

The Foci are on the major axis, so we get $X = \pm c$ and $Y = 0$.

$$\rightarrow x - 3 = \pm 4 \quad \text{and} \quad y + 2 = 0 \quad \rightarrow \quad x = 7, -1 \quad \text{and} \quad y = -2.$$

Foci: $(7, -2)$ and $(-1, -2)$

Sketch????

....

DONE

One axis: $Y = (y + 2) = 0 \quad \therefore \quad y = -2$. Now, if $Y = 0$, then

$$(x - 3)^2 = 5^2 \rightarrow x - 3 = \pm 5 \quad \therefore \quad x = 8, -2$$

Major axis: $y = -2$.

Other axis: $X = (x - 3) = 0 \quad \therefore \quad x = 3$. Now, if $X = 0$, then

$$(y + 2)^2 = 3^2 \rightarrow y + 2 = \pm 3 \quad \therefore \quad y = 1, -5$$

Minor Axis: $x = 3$

$$\text{Center} \left(\frac{(8-2)}{2}, \frac{-2-2}{2} \right) = (3, -2)$$

Foci are always on the major axis where $c = \sqrt{a^2 - b^2} = \sqrt{5^2 - 3^2} = 4$.

$$\text{For Foci, } X = \pm c \quad \text{and} \quad Y = 0 \rightarrow x - 3 = \pm 4 \quad \therefore \quad x = 7, -1.$$

Foci are on the major axis: $y = -2$

Hence, Foci are: $(X, Y) = (\pm c, 0)$

Foci are: $(7, -2)$ and $(-1, -2)$

Standard-Form Equations for Ellipses Centered at the Origin

Foci on the x-axis: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b)$

Center-to-focus distance: $c = \sqrt{a^2 - b^2}$

Foci: $(\pm c, 0)$

Vertices: $(\pm a, 0)$

Foci on the y-axis: $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \quad (a > b)$

Center-to-focus distance: $c = \sqrt{a^2 - b^2}$

Foci: $(0, \pm c)$

Vertices: $(0, \pm a)$

In each case, a is the semimajor axis and b is the semiminor axis.

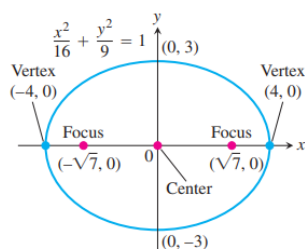


FIGURE 10.8 An ellipse with its major axis horizontal (Example 2).

EXAMPLE 2 Major Axis Horizontal

The ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

(Figure 10.8) has

Semimajor axis: $a = \sqrt{16} = 4$, Semiminor axis: $b = \sqrt{9} = 3$

Center-to-focus distance: $c = \sqrt{16 - 9} = \sqrt{7}$

Foci: $(\pm c, 0) = (\pm\sqrt{7}, 0)$

Vertices: $(\pm a, 0) = (\pm 4, 0)$

Center: $(0, 0)$.

EXAMPLE 3 Major Axis Vertical

The ellipse

$$\frac{x^2}{9} + \frac{y^2}{16} = 1, \quad (6)$$

obtained by interchanging x and y in Equation (5), has its major axis vertical instead of horizontal (Figure 10.9). With a^2 still equal to 16 and b^2 equal to 9, we have

Semimajor axis: $a = \sqrt{16} = 4$, Semiminor axis: $b = \sqrt{9} = 3$

Center-to-focus distance: $c = \sqrt{16 - 9} = \sqrt{7}$

Foci: $(0, \pm c) = (0, \pm\sqrt{7})$

Vertices: $(0, \pm a) = (0, \pm 4)$

Center: $(0, 0)$.

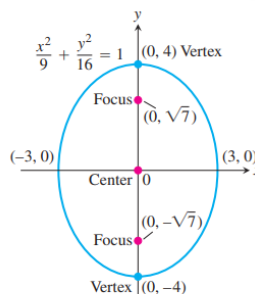
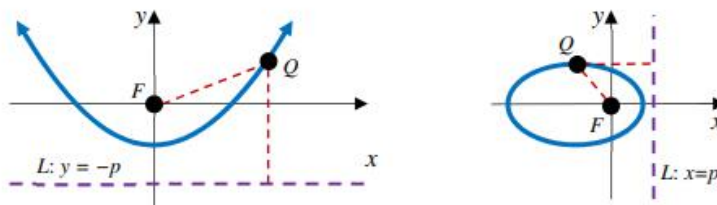


FIGURE 10.9 An ellipse with its major axis vertical (Example 3).

Conic Sections in Polar Coordinates

Conic Sections General Definition

A **conic section** can be defined by placing a fixed point at the origin, $F(0,0)$, called the **focus**, and drawing a line L called the **directrix** at $x = \pm p$ or $y = \pm p$. The conic section is the set of all points $Q(x, y)$ for which the ratio of the distance from Q to F to the distance from Q to the directrix is some positive constant e , called the **eccentricity**. In other words, $\frac{d(Q, F)}{d(Q, L)} = e$.



Warning: the eccentricity, e , is **not** the Euler constant $e \approx 2.71828$ we studied with exponentials

Polar Equation for a Conic Section

A **conic section** with a focus at the origin, **eccentricity** e , and **directrix** at $x = \pm p$ or $y = \pm p$ will have polar equation:

$$r = \frac{ep}{1 \pm e \sin(\theta)} \quad \text{when the directrix is } y = \pm p$$

$$r = \frac{ep}{1 \pm e \cos(\theta)} \quad \text{when the directrix is } x = \pm p$$

Notes: In standard form, conic sections are either symmetric with respect to x –axis (Directrix is vertical) or with respect to y –axis (Directrix is Horizontal).

Focus of the Conic section is at the origin , $F = (0,0)$. Say, $p > 0$.

Directrix	Conic Section
$x = p$	$r = \frac{ep}{1 + e \cos \theta}$
$x = -p$	$r = \frac{ep}{1 - e \cos \theta}$
$y = p$	$r = \frac{ep}{1 + e \sin \theta}$

$y = -p$	$r = \frac{ep}{1 - e \sin \theta}$
----------	------------------------------------

Find the equation of the Conic section with directrix $x = 3$ & eccentricity $e = 2$.

Note: Since $e = 2 > 1$, so it is a hyperbola. Here $p=3$. $E=2$

Directrix is vertical $\rightarrow$ The conic section is perpendicular to x-axis.

Equation: $r = \frac{ep}{1+e \cos \theta} \quad \therefore \quad r = \frac{6}{1+2 \cos \theta}$

Relation Between the Polar Equation of a Conic and its Shape

For a **conic section** with a focus at the origin, **eccentricity** e , and **directrix** at $x = \pm p$ or $y = \pm p$,

when $0 < e < 1$, the graph is an ellipse

when $e = 1$, the graph is a parabola

when $e > 1$, the graph is a hyperbola

Example: What is the shape of the conic section? Find the Directrix.

$$r = \frac{3}{5 - 2 \sin \theta}.$$

Solution: The form of the conic section must be $r = \frac{ep}{1 \pm e \sin \theta}$.

Now, $r = \frac{3}{5-2 \sin \theta} = r = \frac{3}{5\left(1-\frac{2}{5} \sin \theta\right)}$

$$\therefore r = \frac{\frac{3}{5}}{1 - \frac{2}{5} \sin \theta} = \frac{ep}{1 - e \sin \theta}$$

Here, eccentricity: $e = \frac{2}{5}$. Since $0 < e < 1$, so the conic section is an Ellipse.

Also,

$$ep = \frac{3}{5} \rightarrow \frac{2}{5} p = \frac{3}{5} \quad \therefore p = \frac{3}{2}.$$

Directrix: $y = -\frac{3}{2}$.

.....

One of the vertex: Midpoint between the (0,0) and the line $y = -\frac{3}{2}$

The major axis is the y-axis. $x = 0$.

Vertex $(-\frac{3}{4}, 0)$

Important: Graphing Conic sections (parabola and Ellipse) in polar coordinate.

Important Topics of This Section

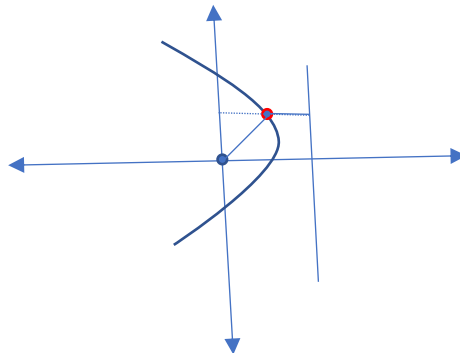
Polar equations for Conic Sections

Eccentricity and Directrix

Determining the shape of a polar conic section

**** Derive the formula for the **parabola** with focus $F = (0,0)$ and directrix $x = p$.

Parabola in Polar coordinates:



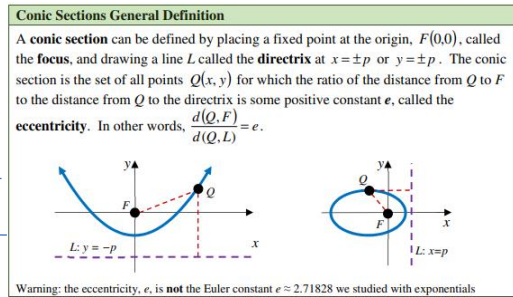
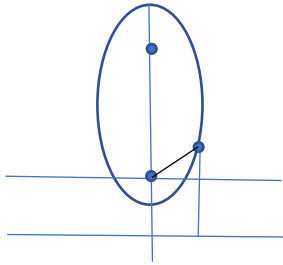
Fixed point Focus $F = (0,0)$ and Fixed line Directrix line L $x = p$.

Consider an arbitrary point $Q(x, y) = (r, \theta)$. By definition

$$\frac{d(Q, F)}{d(Q, L)} = 1 \rightarrow \frac{r}{p - x} = 1 \rightarrow \frac{r}{p - r \cos \theta} = 1$$

$$\rightarrow r = p - r \cos \theta \rightarrow r + r \cos \theta = p$$

$$\therefore r = \frac{p}{1 + \cos \theta}$$



Ellipse: $Q = (x, y)$, Focus $F = (0, 0)$, Directrix Line $L: y = -p$

Parabola: $Q = (x, y)$, Focus $F = (0, 0)$, Directrix Line $L: y = -p$

$$\frac{d(Q, F)}{d(Q, L)} = e ;$$

[$e = 1$ for parabola and for ellipse $0 < e < 1$]

In polar coordinate: $\frac{d(Q, F)}{d(Q, L)} = e \rightarrow \frac{r}{y+p} = e$

$$\rightarrow \frac{r}{r \sin \theta + p} = e \rightarrow r = er \sin \theta + ep$$

$$\rightarrow r - er \sin \theta = ep$$

$$\therefore r = \frac{ep}{1 - e \sin \theta}$$